



Deep learning-based EEG emotion recognition: a comprehensive review

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Abstract

Emotion recognition has been used in a wide range of different fields, such as human–computer interaction, safe driving, education and medical treatment. Compared with text, speech, expression and other physiological signals, electroencephalogram (EEG) signals can reflect an individual’s emotion states more directly, objectively and accurately, and are less affected by the individual’s subjective consciousness. Following the continuous progress of EEG acquisition equipment over recent years, the acquisition of EEG signals has become increasingly convenient. Emotion recognition based on EEG signals has attracted more and more interest from a wide variety of researchers. This paper first reviews the basic theory of emotion recognition, including discrete and continuous emotion models, the development of acquisition equipment and the internal relationship between EEG signals and emotion recognition. A review of the application of deep learning algorithms to EEG emotion recognition is then presented, with a focus on the extraction of deep EEG features and emotion recognition by single deep learning models, attention-based deep learning models, hybrid deep learning models and domain adaptation-based deep learning models. Taking the public datasets SEED, DEAP, SEED-IV, DREAMER and MPED as examples, representative methods of subject-dependent and subject-independent experiments are quantitatively compared and analysed, and the significance of subject-independent research is indicated. Finally, aiming at the problems of dataset’s application limitations, model optimisation and subject variability in existing research, the possible solutions to the corresponding problems are summarised, and the future development trends and prospects are proposed.

Keywords Emotion recognition · Deep learning · EEG · Subject-independent

1 Introduction

‘Letting the machine have emotion’ is one of the research goals of affective computing [1], which was first mentioned in *Affective Computing* [2] by Professor Picard of MIT in 1997, and affective computing has since officially become an important branch of computer science. It is an

interdisciplinary field involving computer science, psychological science, social science, medicine and many other disciplines.

In the field of affective computing, emotion recognition has always been one of the most intensively researched topics, as it can promote human–computer interaction [3] and promote the development of artificial intelligence [4]. The recognition of emotional content can be divided into two areas, direct and indirect emotion recognition, depending on the research object [5]. Direct emotion recognition involves recognising direct performance information from pictures and videos, while indirect emotion recognition involves recognising the stimulus responses of subjects after viewing the content of photos and videos. For indirect or implicit recognition, the subject information that is collected may include facial videos, physiological signals and questionnaire data. Compared with text [6], speech and expressions [7], physiological signals have an advantage they can reflect individual

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emotional states more directly, objectively and accurately, and are less affected by individual subjective consciousness [8]. Of these, EEG data are one of the most widely used signal patterns in physiological signals.

In recent years, there have been notable advancements in the field of EEG emotion recognition, with many studies yielding promising results. Qin et al. [9] and Dadebayev et al. [10] both conducted a comprehensive review of the current developments in EEG signal acquisition and the associated technology of machine learning in the context of EEG emotion recognition. Starting from the entire technical route of EEG emotion recognition, Li et al. [11] provided and summarised several research tutorials for the researchers. Moreover, with the sustained development of deep learning (DL), many DL algorithm models with good results have been applied to the task of emotion recognition based on EEG signals [12]. They have also been applied to more emotion recognition systems based on EEG signals, so that the recognition accuracy of the system is improved [12]. For example, in the subject-dependent experiment of Zhang et al. [13] on the SEED dataset, the recognition accuracy has reached 99.01%. Furthermore, DL models are basically composed of intricate network structures. Some comprise a single neural network, such as convolutional neural networks (CNNs) [14–23], recurrent neural networks (RNNs) [24, 25], graph neural networks (GNNs) [26–28] and so forth. Others are parallel or serial structures of diverse types of networks [29–40]. Additionally, some models amalgamate the learning methods of reinforcement learning or transfer learning into the process of feature learning [41–59]. These EEG emotion recognition models based on DL develop rapidly, and the network structures used are complex and diverse. However, there is still a lack of classification and summary of these methods.

Consequently, in this paper, we first categorise deep learning models into four distinct types according to the overall architecture of deep learning-based EEG emotion recognition model and the type of artificial neural network used. Then, we analyse the subject-dependent and subject-independent experimental results on multiple datasets in terms of different dimensions such as accuracy and stability, especially the DEAP [60] and SEED datasets [61]. At last, it is essential to present the issues and resolutions pertaining to “subject bias”, “model overfitting” and other related topics, as outlined in the existing research findings, providing a clear direction for future research in the field of EEG emotion recognition. In this way, this paper systematically reviews the deep learning-based EEG emotion recognition methods as follows:

- We focus on the application of DL frameworks of EEG emotion recognition. We categorise DL frameworks into four main categories: Single DL models, attention-

based DL models, hybrid DL models, and domain adaptation-based DL models, to carry out a theoretical analysis and give a basic introduction to the latest algorithms in this field

- In the experimental section, we compare the results of different models on classification tasks with subject-dependent and subject-independent experiments on different EEG public datasets, such as DEAP, SEED, SEED-IV, DREAMER and MPED. In this context, particular attention is paid to the practical implications of subject-independent experiments.
- Finally, we present a series of promising solutions that provide practical advice on the main problems encountered in this field, including limitations in the application of datasets, model optimisation and subject variability.

The rest of the review is organised as follows. In Sect. 2, some basic background information related to EEG emotion recognition is introduced, including the definition of an emotion quantification model, the development process of EEG signal acquisition equipment and the specific correlation between EEG and emotion recognition. Then, starting with some DL frameworks, we introduce models that have given good EEG emotion recognition results in recent years, briefly analyse these results and summarise their respective strengths and weaknesses in Sect. 3. In Sect. 4, we introduce some public datasets for EEG emotion recognition and describe the evaluation metrics used to measure the performance of the algorithms. Finally, existing applications and future trends in the development of EEG emotion recognition are briefly introduced, and current problems are analysed in Sect. 5, with a particular focus on improvements to the high tendency of complex models to be overfitted, and the low accuracy of subject-independent emotion recognition results on DEAP and other datasets.

2 Related theories

2.1 Definition of an emotion state

An emotion is a type of psychological state involving various elements, such as cognition, physiology, experience and behaviour. It is an evolutionary feature of organisms that is used to cope with and control the living environment. For humans, an emotion is a qualitative concept that does not need to be accurate, with a specific numerical representation. However, for computers, emotions need to be quantified to facilitate calculations. In the field of emotion recognition, the most widely used quantitative theoretical model is the emotion classification

model, which includes the discrete emotion quantification model and the continuous dimension emotion quantification model.

2.1.1 Discrete emotion quantification models

The discrete type of emotion quantification model divides emotions into independent entities, with no correlations between pairs of emotions, or redefines mixtures of basic emotion types as new emotions. Each of the discrete emotion models proposed so far uses different classification standards and specific emotion classification schemes. Ekman et al. [62] used expression analysis to divide emotions into six basic types: Happiness, anger, disgust, fear, sadness and surprise. This classification method is now widely accepted, and is also the method used for emotion annotation in many public datasets. Ortony et al. [63] proposed the OCC model, one of the most complete emotion models to be developed in this field. It divides 22 basic emotions according to three types of emotion-inducing reasons: Event, subject and object. Lazarus [64] divided emotions into 15 basic types, such as happiness, anxiety and anger, each of which had a corresponding core related theme. Cowie et al. [65] proposed secondary types of emotion by comparing the positive or negative properties of the basic emotion types, meaning that the boundaries of emotional categories were more carefully distinguished, which inspired further research on coarse-grained emotions. Borth et al. [66] chose to use adjective-noun pairs to describe sentiment types in visual content, which were represented using an emotion wheel [67], and this representation enabled advanced sentiment analyses of multimedia content, resulting in a large sentiment concept classifier.

Discrete emotion quantification models are also applicable to a number of scenarios in EEG emotion recognition. For instance, in the SEED [61] acquisition process, subjects viewed three discrete emotion types of positive, neutral and negative videos, and the EEG signals induced by the different video stimuli were recorded. The emotion types of stimulation videos in SEED-IV [68] were differentiated into four types: Happy, sad, fearful and neutral. Additionally, the EEG signals generated during the viewing process were recorded as raw EEG data in real time.

2.1.2 Continuous dimensional type of emotion quantification model

In the continuous dimensional type of emotion quantification model, the different emotions are mapped onto a customised two-dimensional or three-dimensional emotion space, and the quantified emotions are represented by multidimensional vectors. Russell [69] proposed the two-

dimensional affective classification model, also known as the circumplex classification model of affect or valence-arousal (VA) affective model, as shown in Fig. 1. The horizontal axis represents the valence, which reflects the negative or positive degree of the emotion, and the vertical axis represents the arousal, which shows the intensity of the emotion. This model can be applied to regression tasks based on the specific value of the corresponding emotion, and can also be applied to multi-classification tasks according to the relative position in the emotion space. For example, the four poles can represent four discrete classes of emotion, such as happy (positive/high arousal), relaxed (positive/low arousal), bored (negative/low arousal) and scared (negative/high arousal).

For fear and anger, where there is a need to consider the subjective dominance of the individual, the two-dimensional emotion model cannot distinguish these well. In view of this, Mehrabian [70] proposed a three-dimensional space representation of emotion, in which the type of emotion is defined using both axes and poles. Emotions are distributed at different positions between the poles of each axis, and this model is called the valence-arousal-dominance (VAD) model. Cowie et al. [65] proposed the inverted cone three-dimensional emotion model, also known as the 'emotion wheel', which defines three different dimensions: Polarity, similarity and intensity. In this scheme, the basic emotions are combined into new composite emotions, which offer more possibilities for emotion labelling. Fontaine et al. [71] also proposed a four-dimensional emotion model. Using the principal component analysis method, they concluded that in addition to pleasure, dominance and arousal, unpredictability was an important emotional factor, and their approach was equally applicable to different linguistic environments and cultural situations. Cambria et al. [72] proposed a new emotion classification model, called the hourglass emotion model,

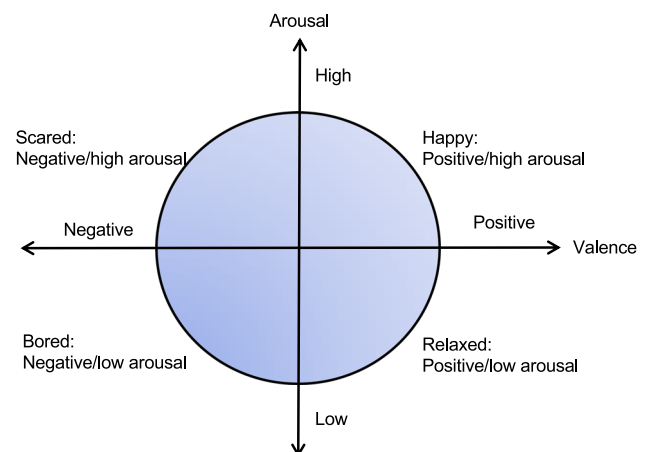


Fig. 1 The circumplex classification model of affect

which was specifically developed for emotional human–computer interaction and emotional analysis. This model represented emotional states through labels and four independent but simultaneous dimensions, with the aim of accurately describing people’s affective experiences. The emotional hourglass can be used in human–computer interaction environments to measure pleasantness, attention, sensitivity and aptitude separately. The overall structure gets its name from the hourglass-like representation of emotion through intensity.

The dimensional type of emotion quantification models can also be divided into two-, four- or eight-class quantification models with labels, depending on the corresponding parameter ranges that are defined. This general method can transform the regression task into a classification task, which is closer to the application model form in real life. Some methods [73, 74] elect to apply the VA model on DEAP [60] and DREAMER [75], with the range of values for both valence and arousal falling between 0 and 9. Valence scores between 1 and 5 are deemed to fall within the negative category, while scores between 5 and 9 are considered to be within the positive category. Similarly, arousal scores between 1 and 5 are classified as low arousal, while scores between 5 and 9 are regarded as high arousal.

2.2 Development of acquisition equipment

Methods of acquiring EEG signals have become simpler and more convenient following the developments in EEG equipment in recent years [9]. Through the continuous efforts of many researchers, the location settings of each EEG electrode (wet and dry) and their respective importance have been studied in detail. Wet electrode devices are widely used in clinical medical applications. Hinrichs et al. [76] used an Inomed PL231 clinical EEG recorder to test the validity of their system. Dry electrodes are most commonly used as EEG acquisition devices, as these can avoid electromagnetic interference while the resting state EEG power and event-related potentials are comparable to those of the wet electrode. Grozea et al. [77] proposed a low-cost metal-coated bristle sensor that mainly detected the α frequency band. Huang et al. [78] proposed a novel type of active comb-shaped dry electrode, which was mainly used to measure EEG signals from the scalp. However, when non-contact dry electrodes are used, the amplitude of the signal is too low to be recorded, and the quality of the collected signal is degraded due to the distance. Semi-dry electrodes have therefore been designed that can combine the advantages of wet and dry electrodes and effectively avoid their disadvantages, and only tens of millilitres of electrolyte are needed to obtain relatively stable EEG signals.

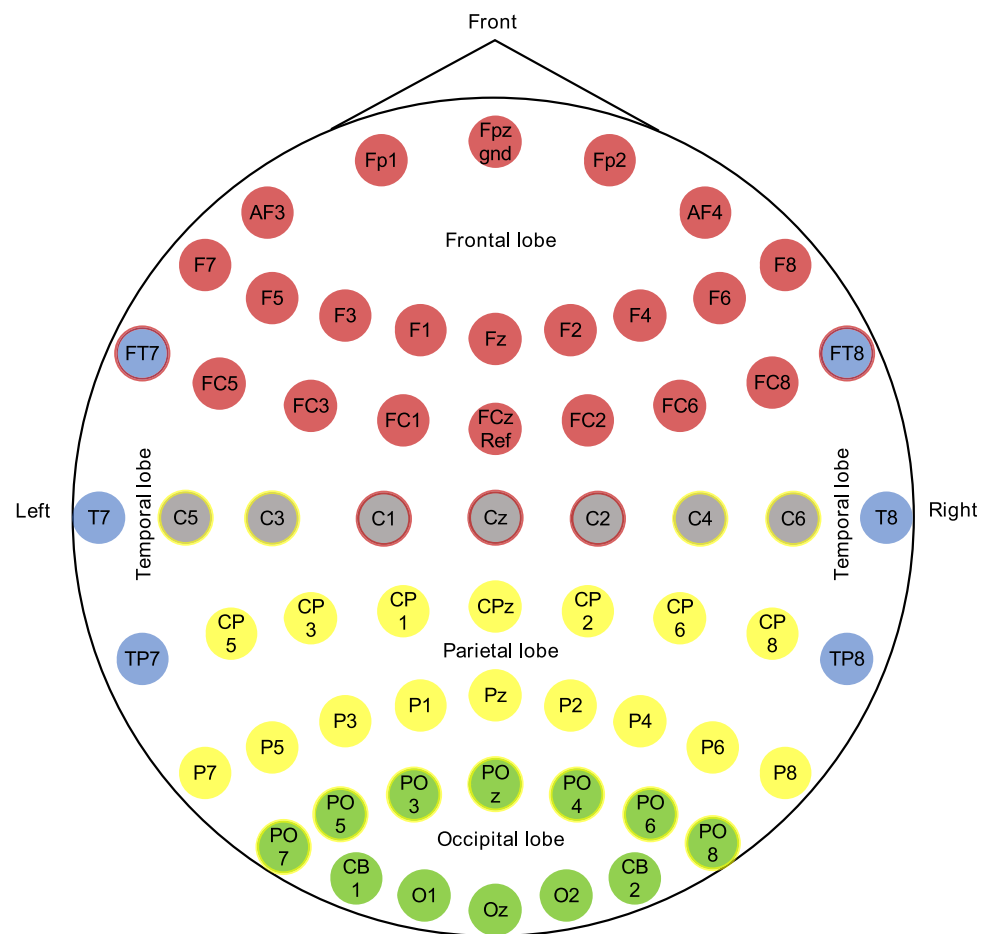
Hairston et al. [79] compared four kinds of EEG equipment through experiments. Zheng et al. [68] reported that EEG information collected via only six key electrodes could give high recognition accuracy, providing a theoretical basis for the development of real-time EEG detection systems in future. A significant proportion of currently available EEG acquisition devices, particularly those designed for use as wearable technology, are employed in a range of commercial applications. Notable manufacturers of consumer-grade EEG devices include open brain–computer interface (OpenBCI) [80], Emotiv [81], InteraXon [82] and NeuroSky [83]. The devices in question are employed extensively in gaming, entertainment and health-related contexts, facilitating user interaction and control of applications through the use of brainwave activity. Zhao et al. [84] has shown that it is acceptable and necessary to introduce a short-term calibration stage before the real-time recognition starts. Varieties of EEG devices and tools are developed, providing researchers, developers and consumers with a comprehensive range of options for exploring and utilising brain–computer interface technology. The specific products and product parameters are detailed in Table 1.

2.3 Relation between EEG and emotion recognition

Emotion is mainly controlled by the brain’s autonomic nervous system. Study has shown that various brain regions and the connections between them have different effects on emotions [85]. In this paper, using an updated activation likelihood estimation (ETA) method of meta-analysis to analyse and explore the whole brain, it has been concluded that certain brain regions respond more strongly to facial expressions. In addition to the higher activation of facial emotion in the fusiform face area (FFA) and its involvement in facial expression processing, regions such as the amygdala and parahippocampal gyrus also play an essential role in facial expression perception, and are always involved in processing facial expressions. Li et al. [73] proposed that multichannel and spatial information from EEG signals should be used to consider the correlations between different electrodes and different brain regions, such as the connections between the frontal lobe, temporal lobe, parietal lobe, occipital lobe and central sulcus and the influence of salience on emotion. Figure 2 shows location information for the 62 brain electrodes of an EEG and the brain region to which each electrode belongs. The red area represents the frontal lobe, blue represents the temporal lobe, yellow the parietal lobe and green the occipital lobe. Meanwhile, ‘Fp’ stands for ‘frontal poel’, ‘A’ for ‘anterior’, ‘F’ for ‘frontal’, ‘C’ for ‘central’, ‘T’ for ‘temporal’,

Table 1 Introduction to affordable EEG devices

Device	Company	Year	Count	Electrodes	Sampling rate (Hz)
Muse	InteraXon	2016	4	AF7, AF8, TP9, TP10	256
Ultracortex	OpenBCI	2017	8/16	8: Fp1, Fp2, C3, C4, P7, P8, O1, O2	250/125
Mark IV				16: Fp1, Fp2, F3, F4, F7, F8, T7, T8, C3, C4, P7, P8, P3, P4, O1, O2	
EPOC Flex	Emotiv	2018	32	Configurable on standard 72 channel international 10–20 locations	128
MindWave	NeuroSky	2018	1	Fp1	512
EPOX X	Emotiv	2020	14	AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4	128/256
Insight 2.0	Emotiv	2022	5	AF3, AF4, T7, T8, Pz	128

Fig. 2 The EEG cap layout for 62 electrodes

'P' for 'parietal', 'O' for 'occipital', 'CB' for 'cerebellar' and 'Z' for 'zero'.

When various emotional stimuli are presented, in addition to the different degrees of activation of brain regions and different activation routes of brain electrodes, the pre-processed EEG frequency bands also show different characteristics. Wang et al. [86] concluded that it was most appropriate to use the alpha band to study the relationship between EEG data and emotional changes, and noted that this is the band most frequently seen in the brain's

electrical signals. It typically appears during periods of wakefulness and when people are performing simple response activities. Alpha oscillations have relatively stable spatial patterns between subjects, and are also considered to be a marker of emotional processing. However, Sun et al. [87] proved through experiments that the beta frequency band also shows a significant response to emotional changes; in particular, when subjects are experiencing negative emotions, the beta frequency band at the prefrontal cortex gives a significant response. Zhang et al.

[88] also concluded that each frequency band was related to a change of emotional stimulus based on experimental results, and the delta, alpha and beta bands had the highest correlations with emotion. Fluctuations in the theta frequency band spanned almost all regions of the brain, showing strong spatial connectivity. Choog et al. [89] proved through a correlation coefficient analysis that the beta and gamma bands had strong correlations with anger, fear, surprise and other negative emotions, and that these gradually decreased from the alpha to gamma band. In the field of EEG emotion research, pre-processed EEG signals are usually divided into five main frequency bands that have different characteristics, as shown in Table 2.

When a subject is watching a video, for different video frames, EEG signal fluctuation changes are also different. This proves that it is meaningful to extract EEG feature information by video stimulation. Lew et al. [90] used the changes in EEG signals to select key frames from a video to carry out summarisation of the video, and introduced a time synchronisation module to align the modal information to compensate for the modal differences between the video and the EEG signals. An attention mechanism was used to transform the EEG representation into the visual representation space. This method made it possible to obtain individual emotions from EEG signals, and could be applied to multimedia content-related visual tasks. It provided a new idea for research into combinations of direct and indirect emotion recognition, as well as fusions between different modalities. So, it is necessary to directly use EEG to study the mechanism of emotion cognition.

The basic research process of EEG emotion recognition based on DL are shown in Fig. 3, including EEG acquisition, signal pre-processing, feature extraction and feature learning. The models used in each of the four processes are widely different, and a variety of brain stimulation materials are used. In summary, the whole process is to apply DL models to extract and learn emotion-related EEG features to distinguish between different emotional states.

Firstly, videos, pictures and audio are typically shown to the subjects as stimulation sources for emotional signals, and the content is changed to induce different emotional data for researchers to collect. EEG signal pre-processing is a very important basic step. It is also necessary to pre-process the original EEG signals, for example to remove artefacts and reduce noise, in order to identify and select the EEG activities related to emotion. Interference can be removed by bandpass filtering, and the signal artefacts can be removed by independent component analysis (ICA) [91] or other methods. As the input to the DL network framework, the EEG signal can be used after appropriate processing. The most commonly used frequency domain features are the power spectral density (PSD) [92] and differential entropy (DE) [93], although there are others such as time domain and spatial domain features. The DE formula is expressed as follows:

$$\text{PSD} = - \int_x p(x) \log(p(x)) dx \quad (1)$$

where $p(x)$ is the probability density function. In addition, the DE features can also be calculated using the PSD:

$$\text{DE} = \frac{1}{2} \log(\text{PSD}) \quad (2)$$

Then, the feature information can then be extracted by a windowing operation to obtain better quality signals and their features for learning. Some representative DL methods include CNNs [14–23], RNNs [24, 25] GNNs [26–28] and Transformers, which can give excellent results after feature learning. Finally, the extracted features are input to the classifier for classification, and discrete sentiment types corresponding to the samples (or the fitting values in regression tasks) are obtained.

Table 2 Introduction to EEG frequency bands

Frequency bands: (Hz)	Introduction (related to emotion processing)
δ : 1 ~ 3	Usually occurs during slow-wave sleep, involved in motivational processes and implicated in the brain reward system
θ : 4 ~ 7	Usually occurs in older children and adults when they are drowsy or awake, involved in the discrimination of emotional stimuli
α : 8 ~ 12	Occurs when people perform simple reaction activities or when people close their eyes and rest quietly, associated with human attention
β : 13 ~ 30	Generally occurs when the person is alert, active, anxious or focused
γ : 31 ~ 50	Occurs mostly in the central nervous system, also sensitive to negative emotions

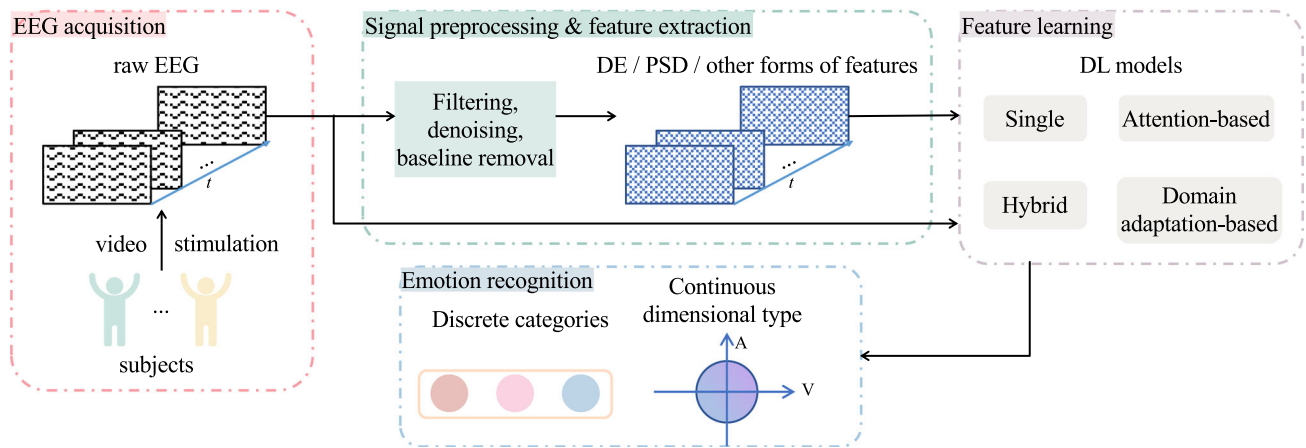


Fig. 3 Basic research methodologies of EEG emotion recognition

3 Deep learning methods for EEG emotion recognition

With the development of DL, increasing numbers of algorithms based on DL have been applied to tasks related to EEG emotion recognition. However, different types of network frameworks and different classification methods have particular advantages and disadvantages. In this section, we categorise DL models into four groups: Single, attention-based, hybrid and domain adaptation-based DL models. The proportion of summarised DL models applied to EEG emotion recognition is shown in Fig. 4.

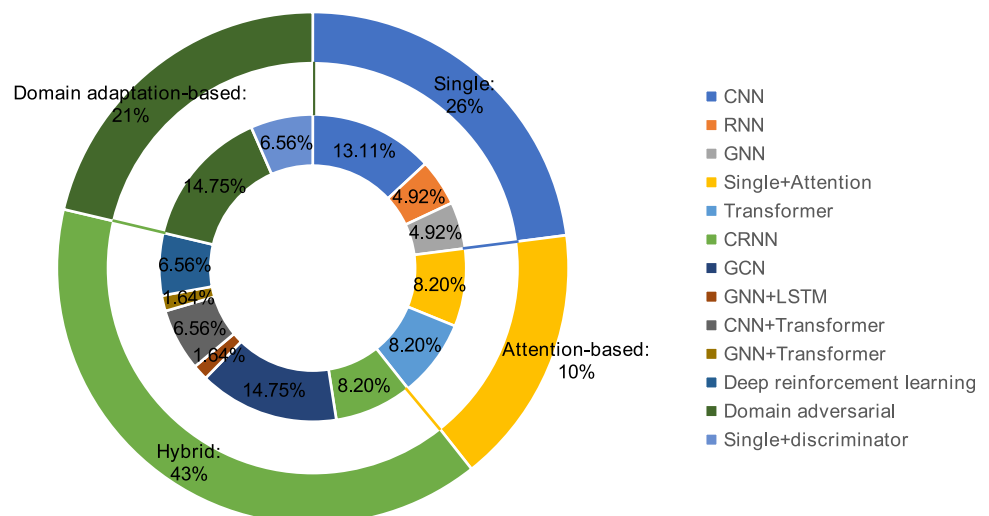
As illustrated in the figure, hybrid models account for the largest proportion of 43%, comprising six distinct models: Convolutional recurrent neural network (CRNN), graph convolution network (GCN), GNN with LSTM, CNN with Transformer, GNN with Transformer and deep reinforcement learning (DRL). The single models account for a total of 26%, with CNNs representing the largest

category at 13.11%. Domain adversarial and attention-based models accounted for 21% and 10%, respectively. In the subclassification category, domain adversarial and GCN both accounted for the largest proportion at 14.75%.

3.1 Single DL models

Single DL models are structures containing only one basic network framework, and can be used for feature extraction, feature learning and classification. These simple logical structures can clearly represent the dimensions and connotations of the input features. However, the application of basic neural networks to EEG emotion recognition has certain limitations, as the dimensions of EEG data are different from those of commonly used models, and the input dimensions of the data need to be adjusted. CNNs, RNNs and GNNs are three representative types of models, and their basic structures are shown in Fig. 5.

Fig. 4 The proportion of DL models



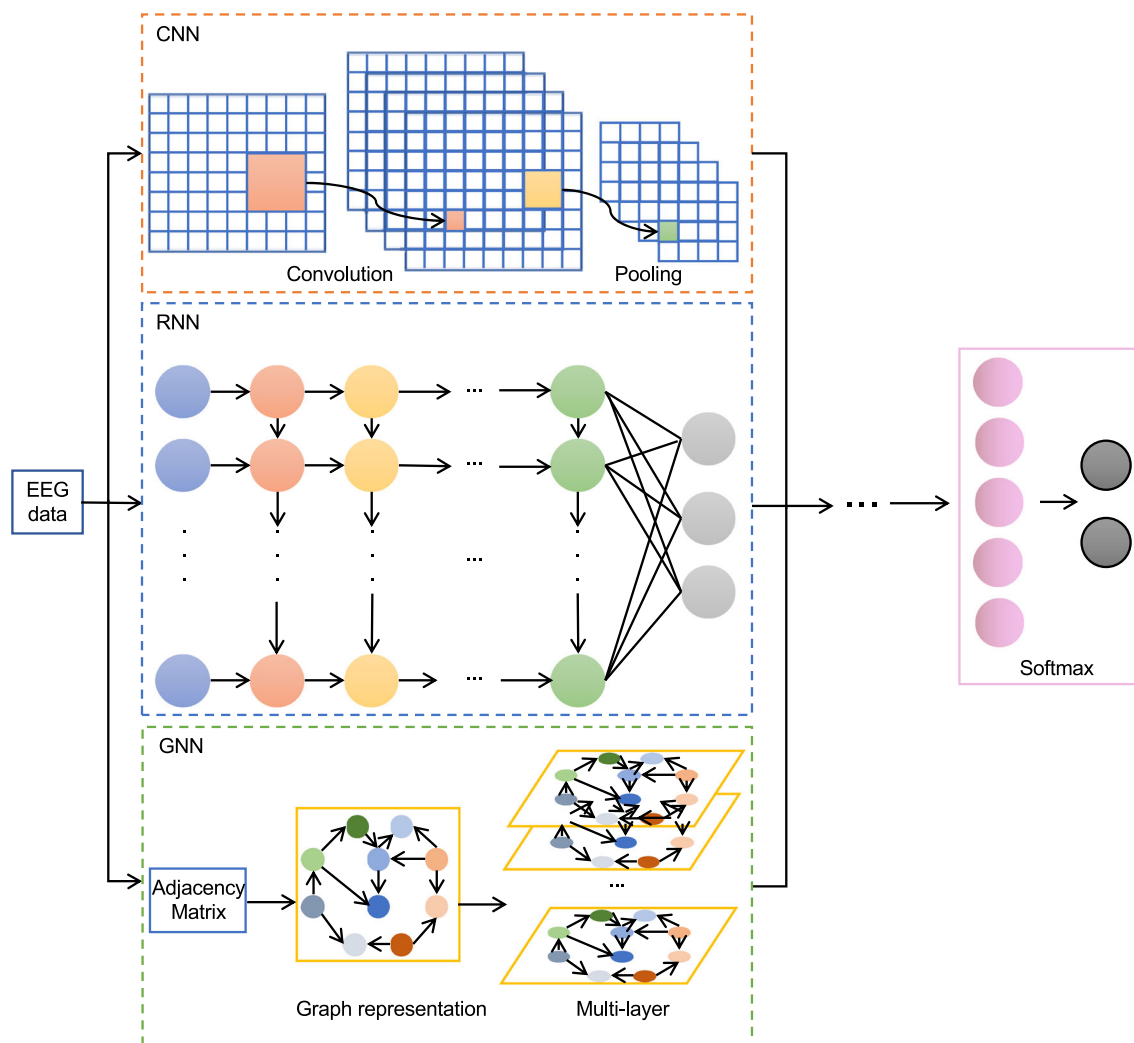


Fig. 5 The basic structures of single DL models

CNNs are good at modelling local receptive field information, but pay less attention to global information, meaning that most convolution-based methods more or less ignore the transmission characteristics and spatio-temporal correlations between the different EEG electrodes. Tripathi et al. [14] applied a CNN model to the task of EEG emotion classification for the first time and compared the results with those of a deep neural network (DNN), thus proving the effectiveness of a CNN for EEG emotion research. Following this study, CNN-based optimisation models have increasingly been used in research on emotion recognition using the EEG modality. Since EEG features have varying spatial and temporal characteristics, different CNN models can be selected according to the different dimension transformations. A two-dimensional convolution is widely used.

Li et al. [15] proposed a hierarchical convolutional neural network (HCNN), in which the differential entropy features of different channels were organised into two-

dimensional spatial maps to classify positive, neutral and negative emotional states. Deng et al. [16] proposed a spatial folding ensemble network (SFE-Net) to extract spatial and temporal information from EEG data; these authors completed the channel information through the use of an improved interpolation algorithm, and processed the spatio-temporal features by combining a 3D-CNN [17]. Yang et al. [18] used a temporal convolutional network (TCN) [19] to model time series information. This approach overcame the problem of gradient explosion and gradient disappearance caused by traditional convolution in the training process, and made the TCN more suitable for the task of EEG emotional recognition by improving the loss function called Additional Margin softmax loss (AM-softmax). AM-softmax is defined as follows:

$$L_{ams} = \frac{1}{N} \sum_i -\log \left(\frac{e^{s \cdot (\cos(\theta_{y_i,i}) - m)}}{e^{s \cdot (\cos(\theta_{y_i,i}) - m)} + \sum_{j \neq y_i} e^{s \cdot \cos \theta_{j,i}}} \right) \quad (3)$$

where N is the number of training samples, y_i is the i th ground-truth class, θ is the angle and m and s are hyperparameters.

There are many other convolutional models that have been optimised through the use of multidimensional channels, frequency selection, spatial topology and temporal continuity of EEG signals, especially for the reorganisation of feature structures. Wen et al. [20] used the Pearson correlation coefficient to reorganise the channel information to extract the degree of association between different electrodes. Yang et al. [21] started from the frequency domain and channel of the EEG modes, and changed the matrix dimensions of the input features of the model, which reduced the complexity of the multidimensional channel operation. Based on a consideration of time and frequency, Cui et al. [22] devised a special treatment for the asymmetry of EEG spatial regions. Rudakov et al. [23] proposed a multitask convolutional neural network in which the PSD and DE features were selected, and the frequency information was combined into a three-dimensional feature vector of size $8 \times 8 \times 9$ by combining the topological structure of the EEG electrode space dimensions. A fully convolutional network structure was used, but this approach still achieved excellent performance, 96.28% for valence and 96.62% for arousal on the DEAP dataset. This study showed that EEG feature extraction and feature selection are extremely important; if the main information of the channel, space and time of EEG features can be retained as far as possible in the process of feature extraction, the recognition results can be accurate even when only a series of convolution operations is used.

An RNN can capture time series information, and in particular, a long short-term memory (LSTM) module can encode the walking path caused by emotional fluctuations between different EEG electrodes and indicate the dependence between electrodes. Zhang et al. [94] proposed a spatial-temporal RNN (STRNN) in which spatial and temporal dependence information was extracted in the spatial RNN and temporal RNN layers, respectively, and sparse projection was performed on the hidden states to improve the recognition ability of the model. This network could not only recognise the emotional types of EEG modalities, but also gave good results on the task of facial emotion recognition, 95.4% on CK+ [95]. Zhang et al. [24] were the first to introduce a path mechanism into the field of EEG emotion tasks, and encoded the dependencies between nodes through LSTM to obtain a path embedding representation. Li et al. [25] divided the EEG space into

left and right parts, and used four RNN modules to capture the differences and connections between the bidirectional and bilateral hemispheres from the horizontal and vertical directions, respectively. However, the average results of this method in a subject-independent experiment, 85.40% on SEED, were not as good as those of a subject-dependent experiment, 93.12% on SEED, which may have been due to the excessive attention paid to spatial information and the neglect of the inherent differences between different samples.

A GNN can fully consider the channel, spatial and regional characteristics of EEG signals, and the different relationships between nodes can be described by constructing different static or dynamic adjacency matrices. A weighted directed graph can be represented in this form:

$$G = \{V, E, A\} \quad (4)$$

where V represents the set of graph nodes, E represents the set of edges and $A \in \mathbb{R}^{n \times n}$ represents the adjacency matrix whose element $A_{ij} > 0$ characterise the relation from node v_i to v_j .

Since EEG features are non-Euclidean data, GNN-based EEG emotion recognition has been developed rapidly. Li et al. [26] studied the representation of an adjacency matrix. Xue et al. [27] used EEG spatial topological information to construct channel-level and region-level graphs. Zhong et al. [28] proposed the regularised graph neural network (RGNN). Two regularisation methods were proposed, namely node-wise domain adversarial training and emotion-aware distribution learning, which could be used to deal with EEG differences between subjects and noisy labels, and were shown to extract more focused discriminative features. In general, methods based on GNN are still of great significance for the task of EEG emotion recognition.

Single DL models can extract and learn EEG features through a major network architecture, meaning that the structures of the overall models are simple and clear, and they are easy to transfer and apply. However, they have many shortcomings. Firstly, a network architecture tends to focus on only one feature dimension, for example CNNs and GNNs tend to focus on spatial information, while RNNs focus on temporal information. Secondly, different structures extract different levels of information: A GNN obtains a global space by constructing a graph representation, while a CNN first learns shallow information and then extracts deep information. Due to the limitations on the step sizes of the convolutional kernel, only a local receptive field can be obtained. Thirdly, EEG features are sequences that include not only channel information, but also information on the frequency, space and time dimensions. Hence, one type of neural network is not sufficient to extract these multidimensional features, and the

correlations between different brain regions and the differences in activation paths between the left and right brain hemispheres also need to be explored by different modules.

3.2 Attention-based DL models

Attention-based DL models can be divided into two types: One involves adding an attention module to a single DL model, while the other is based on a transformer, which improves the mutual correlation between nodes in the feature learning process through the use of a self-attention mechanism.

Adding attention modules to single DL models is one of the most common approaches to optimisation. The addition of functional attention-based modules can make up for the deficiency of more targeted methods by extracting the deep feature information of samples and paying more attention to the connections between channels, frequency and spatio-temporal information to improve the effects of EEG feature learning. Yan et al. [96] proposed a one-dimensional convolutional neural network (1D-CNN) model based on the attention mechanism of frequency bands and brain regions. This approach addressed the issue of the large number of model training parameters, requiring approximately 1/396 and 1/913 of the number of parameters of 2D-CNN and 3D-CNN models, respectively. Furthermore, it facilitated the interpretability for the parameter selection and convolutional operations of the model. Guo et al. [97] raise a functional connectivity-enhanced feature-grouped attention network (FC-FAN) for cross-subject emotion recognition. Chai et al. [98] considered optimisation from the perspective of channel and frequency, and divided the EEG signal into five frequency bands to extract the corresponding DE features. Song et al. [99] augmented the LSTM model with an attention mechanism to facilitate the recognition of deep feature information extracted by three-layer 1×1 convolutional kernels. The method demonstrated superior performance in recognising emotions from physiological signals in the multimodal physiological emotional database (MPED), achieving an accuracy of 72.93% in a triple classification task, outperforming previous methods.

In addition to using a network structure based on graphs, which is conducive to EEG emotion research, a graph attention mechanism can also effectively fuse the weights between different channels, combine the attention unit with the graph structure, improve invariance to fluctuations in emotional intensity and effectively generate representative feature maps. Inspired by the attention mechanism, Song et al. [100] proposed a variational instance adaptive graph method that took into consideration the differences between individuals and the dynamic uncertainty between different EEG signal regions. The authors designed a

coarse-grained graph structure with sparse constraints to make the model more robust, which can be written as:

$$Z_p = \frac{1}{n_{te} - n_{ts}} \sum_{l=n_{ts}}^{n_{te}} \mathbf{H}_{VIAI}, (p = 1, 2, \dots, n') \quad (5)$$

where Z represents the nodes after graph coarsening, and the p th group includes nodes from number n_{ts} to number n_{te} , $\mathbf{H}_{VIA} \in \mathbb{R}^{n \times 2d'}$ is the concatenated feature, which contains the result from deterministic graph and the result from probabilistic graph.

Transformer-based methods can focus on global spatial information at the same time, and can extract long-distance functional connectivity. The self-attention mechanism in the Transformer architecture is insensitive to the positional information of the input sequence, and it is therefore necessary to include position embedding to build spatial relationships between input elements. In addition, due to the different impedances of different brain regions, there may be a small error in the time between the display of the EEG signal by the acquisition device and the real EEG signal; that is, different EEG electrodes will cause different delays in the EEG signal. Hence, it is also important to grasp the overall time series, not only for the information processing in a short time, but also to pay attention to the signal correlations over long time periods and between short-time periods. In past years, researchers have proposed many advanced methods based on Transformer-based model architectures for EEG emotion recognition tasks, in which attention-based modules have been added to the baseline network. Arjun et al. [101] applied the original vision transformer (ViT) model to EEG classification tasks, which showed that the feature information extracted by selecting shorter sample segmentation times was better, and values of 6 s and 15 s were best. Lu et al. [102] adopted a dual-branch ViT model structure in which one branch was used to extract EEG spatio-temporal features, with the non-overlapping window set to 1 s, while the other branch processed the data in a similar way to the CNN model in order to extract the spatial frequency features of the five different frequency bands. This approach achieved excellent recognition results, 97.55% on SEED and 88.08% on SEED-IV. Wang et al. [103] divided the EEG space into the electrode level and the brain region level, and captured the relationships between electrodes and regions for the remote dependence space. Wei et al. [104] mapped channels onto capsules and characterised them as spatial relationships between features. The capsule network was added at the end of the network to refine and process the emotional state to improve the accuracy, resulting in a 0.47% and 0.41% higher average accuracy on DEAP and DREAMER than the Transformer model, respectively. Liu et al. [105] proposed a spatio-temporal Transformer. In

addition to a spatial attention mechanism for 1 s segmented samples, it also included temporal attention for a sample segment, as well as temporal spatio-temporal and synchronous spatio-temporal attention. Compared with traditional CNNs and LSTM-based models, the main reasons for the excellent performance of Transformer-based models are firstly that an attention mechanism can capture more global spatio-temporal information; secondly, for similar tasks, a Transformer has faster computing performance and stronger real-time performance than other neural networks; and thirdly, it quantifies the time period information, which is conducive to directly extracting the original EEG signal features, which gives high attention scores and enables an end-to-end model to be implemented.

The addition of an attention mechanism is a relatively simple method of optimising a model due to the pluggability of this module. Interaction between the 62 electrodes of an EEG can be realised by adding channel-level attention, and the associations between brain regions can be optimised by adding spatial attention. Self-attention and temporal attention can be used to optimise temporal feature information. In addition, since the Transformer-based model is a self-encoding and decoding model, the order of dimension extraction between modules can be exchanged, and optimal module placement positions and combination methods can be obtained through ablation experiments.

3.3 Hybrid DL models

A hybrid DL model integrates two or more network architectures into a model framework. This type of model fusion can combine the advantages of different models when applying them to the task of recognition. There are several kinds of model combination structures, such as CRNN, GCN, GNN with LSTM, CNN with Transformer, GNN with Transformer and DRL. Of these, the CRNN and GCN are the two that have been most widely studied in recent years [29–40].

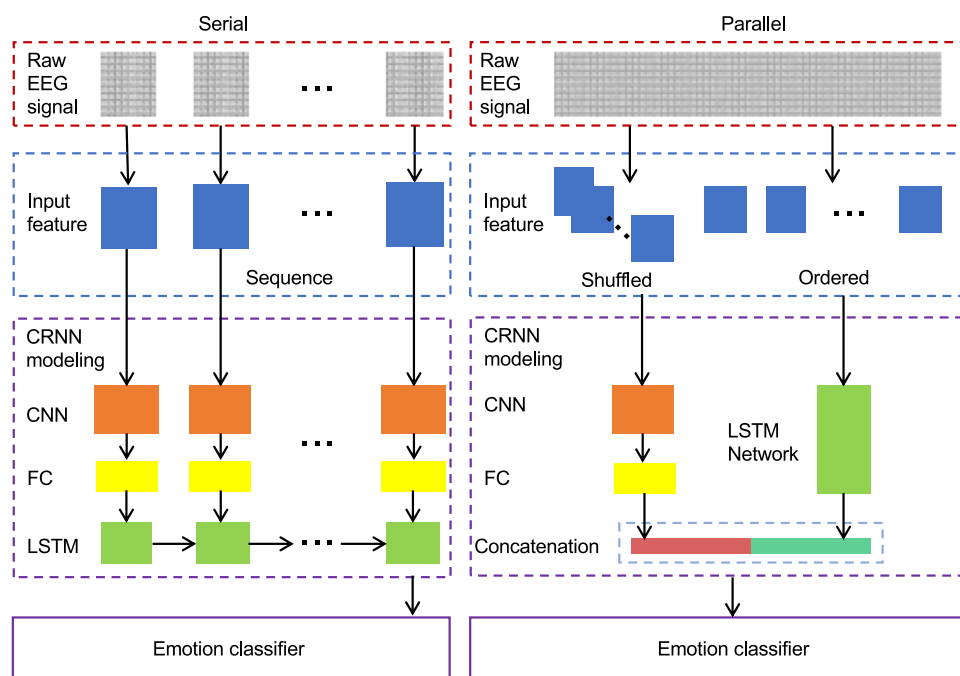
The convolution operation of a CNN can extract the local spatial information of the features, whereas the RNN operation considers the time characteristics of EEG signals, although it ignores the spatial relationship between EEG electrodes and the parallel computing efficiency is low. If CNN and a RNN can be combined, the time series association can be extracted while retaining part of the spatial information, and the spatio-temporal characteristics of EEG can be comprehensively extracted. The two types of CNN and RNN combinations are parallel and serial, as shown in Fig. 6.

In view of this, Yang et al. [29] combined a CNN with an RNN to identify multichannel EEG emotion, converted the chain EEG sequence into a two-dimensional spatial electrode matrix sequence (that is, converted one-

dimensional signals into two-dimensional frames), and used a CNN to learn the channel correlations between physically adjacent EEG signals, with an LSTM to mine context information. Their experiments proved that the accuracy obtained from the pre-processed features was much higher than for the original data information without transformation, improving by nearly 33% and 30% both on valence and arousal recognition task. Kim et al. [30] combined two-layer LSTM model with one-layer attention and two-layer CNN. Tao et al. [31] proposed an attention-based convolutional recurrent neural network (ACRNN), in which an attention mechanism was added to a CNN and an LSTM. The addition of channel attention to the spatial information extraction part and self-attention to the temporal information extraction part enables more discriminative features to be obtained, as a neural network model based on CRNN can extract and combine the spatial and temporal feature information at the same time. This approach also has high inclusiveness for the dimensional input of multidimensional features, the four-dimensional feature representation of EEG signals can contain a lot of information, and more deep discriminative features can be retained through the feature learning of the CRNN model with a high recognition rate, 93.72% for valence and 93.38% for arousal on DEAP, and 97.93% for valence, 97.98% for arousal and 98.23% for dominance on DREAMER. Shen et al. [32] integrated the frequency, spatial and temporal information in EEG features using a combination of a CNN and an LSTM. Xiao et al. [33] integrated PSD into a four-dimensional representation on the basis of extracting differential entropy features. Unlike in the above methods, a sparse representation was selected for two-dimensional mapping of the EEG electrodes, and a two-dimensional representation of size 8×9 was extended to a sparse matrix of size 19×19 . However, a four-dimensional representation contains more information, and a sparse representation was therefore able to improve the efficiency of feature learning. The importance of the electrodes could clearly be observed in the visualisation experiment.

For the sequential problem of time, the graph structure cannot be fully considered, and it can be combined with convolution to extract the correlation information within the channel and between short-distance electrodes. Although a 1D convolution can collect some of the short-time sequence information through a dimension transformation, it cannot effectively learn feature information with a large time span. A GNN usually gives good results when applied to EEG emotion recognition. As a CNN focuses solely on the local spatial features of the brain network, it leads to a loss of high-dimensional spatial patterns, and the use of a kernel with constant size in the convolution operation partly destroys the spatial correlation of the EEG

Fig. 6 Serial and parallel structures of CRNN



signals. However, the characteristic information of EEG signals can be represented by a graph structure, which can preserve the deep internal structure, and the simplified GCN can be expressed as

$$Z = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X \Theta) \quad (6)$$

where $\tilde{A}$ is the sum of n -dimensional degree identity matrix and adjacency matrix of the graph, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, Θ is a trainable weight matrix.

In particular, a graph convolutional neural network (GCNN) combines the advantages of the CNN and a graph structure, and can efficiently capture more discriminative features. The static graph convolution proposed by Wang et al. [34] and the dynamic graph convolution neural network (DGCNN) proposed by Song et al. [106], illustrated in Fig. 7, have been found to demonstrate high levels of recognition accuracy in EEG emotion recognition tasks, with DGCNN achieving 81.91% accuracy on SEED.

Zhou et al. [35] combined static and dynamic graph convolutions to form a progressive graph convolution network (PGCN), in which a two-headed structure was adopted to gradually learn more discriminative EEG features from coarse-grained (easy) to fine-grained (hard). Zhang et al. [13] proposed a sparse dynamic graph convolution, and mainly from a regional perspective, added sparse constraints to the graph structure and optimised it on the basis of dynamic graph convolution. Sun et al. [87] proposed a two-branch dynamic graph convolution with 1456 parameters. These authors reported that the DEAP dataset, commonly used as a reference, yielded the best

recognition results for the extracted PSD features [60], and the SEED dataset gave the best results for the DE features [36]. The DE features calculated by an inverse fast Fourier transform (IFFT) and the PSD features calculated by a short-time Fourier transform (STFT) were obtained by dynamic graph convolution, and the features to be processed and learned were obtained by channel-wise multiplication and concatenation with the dynamic adjacency matrix, which represented the functional connectivity of the electrodes. Qiu et al. [37] applied the idea of residual connection to the structure of a GCN using two-layer multi-head residual graph convolutional network (MRGCN). Liu et al. [38] proposed a global-to-local feature aggregation network (GLFANet) to obtain the overall and regional depth information of an EEG. Lin et al. [39] proposed an improved graph neural network based on channel selection, which combined a one-dimensional convolution with graph convolution. Unlike the methods discussed above, the channel information of the sequence signal was extracted via a one-dimensional convolution, and a graph structure was then used to represent the spatial and functional connections between the different channels after channel selection. Their experimental results showed that the accuracy of acquisition was highest when all of the channel features were selected for feature learning, 90.22% on SEED using all bands, which indicated that in the process of emotion recognition, any collected EEG electrode signal will have different degrees of fluctuation response to emotional changes. Jin et al. [40] proposed the pyramidal graph convolutional network (PGCN), with 12.29M parameters and 12.43M computation, which

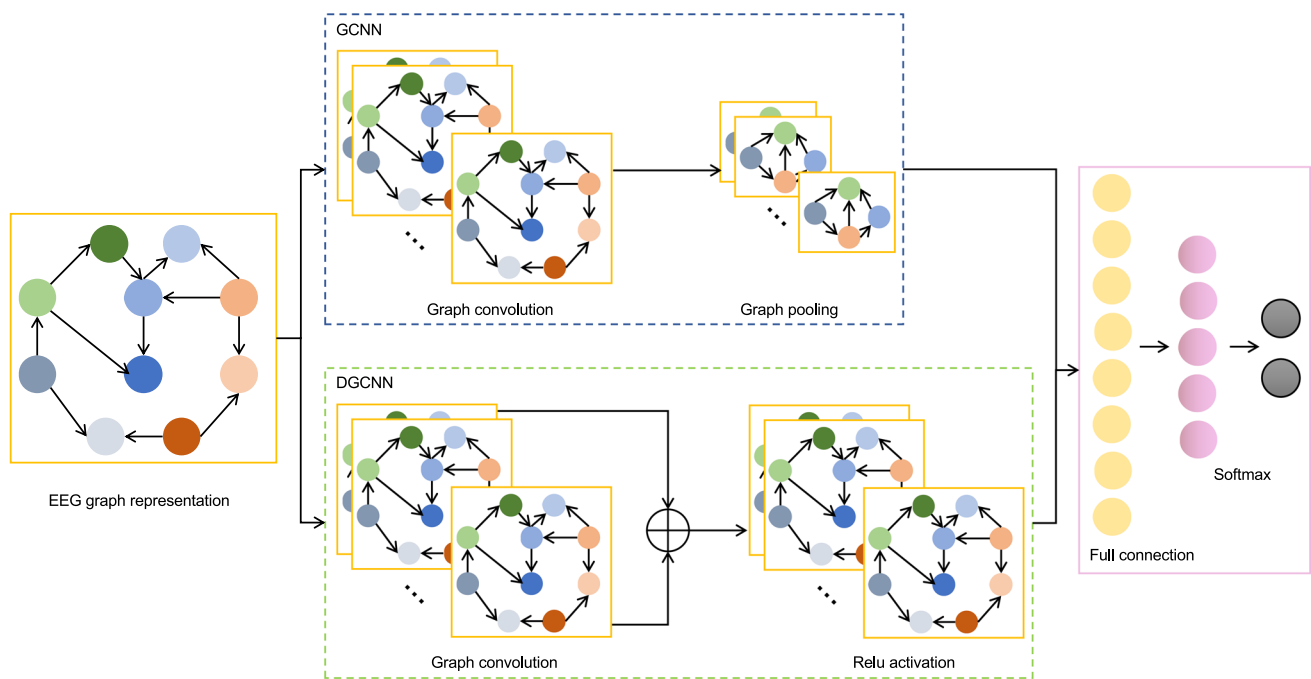


Fig. 7 Structures of GCNN and DGCNN

aggregates features at three levels: Local, mesoscopic and global.

The integration of GNN with LSTM enables both the global spatial features to be preserved and the long-distance temporal interactions to be learned. Liu et al. [107] proposed a network framework that combined sparse graph attention with LSTM. Through the use of a dual-branch structure, the global features containing the intrinsic correlations between different channels and the effective features that added attention to specific channels on the global features were extracted, and the information obtained by their element-wise product was input into the LSTM to predict the emotion results. This method was able to effectively enhance the role of the channel information, which is tightly related to emotional fluctuations, and reduced the role of the channel information, which does not respond strongly to emotional changes.

To optimise the application of the Transformer model, it is necessary to add channel information, consider the dependence between channels, extract global spatial information from different channels and effectively fuse the related information between channels to avoid excessive dependence on the explicit spatial location distribution. In addition to adding and integrating modules related to the attention mechanism, replacing or combining part of the Transformer module with convolution operations is also a commonly used optimisation method. Gong et al. [108] chose an attention-based approach that integrated spatio-temporal DE or original features to extract local spatial and frequency features by convolution. Guo et al.

[109] input the extracted DE features into a deep convolution, and then through an encoder. This method also gave good results in subject-independent experiments, 83.03% on SEED, and the authors discussed the significance of selecting different channels as input to the classifier. Song et al. [110] applied a classic conformer model to the task of EEG emotion recognition, and added a new class activation map to generate a new visual brain map. Sun et al. [111] reconstructed the EEG structure by extracting parallel channel information and reconstructing sequence position mapping. The re-extracted features were input into a 3D-CNN and Transformer, and the spatio-temporal information extracted by the neural network was refused.

A graph neural network can also obtain spatial feature information, and can therefore perform well with a Transformer encoder. Li et al. [112] combined a graph structure with a Transformer as follows. The signal containing channel and time information was first input into the Transformer learning block, and a spatio-temporal attention mechanism was then added to the output results to obtain multiple learnable matrices to describe the graph structure. Their method used leave-one-subject-out (LOSO) for training, and was found to outperform previous methods, 90.37% on SEED and 76.43% on SEED-IV.

DRL combines the perceptual ability of DL with the decision-making ability of reinforcement learning. The biggest advantage of DRL is its ability to learn from interaction, and the paradigm of reinforcement learning is very similar to the process of a human accepting and learning new knowledge to adapt to the surrounding

environment. Methods based on DRL are therefore very suitable for the task of emotion recognition based on EEG signals, as they can learn the activated brain regions of the human brain when experiencing emotion and extract key information from different responses. In a study by Dutta et al. [41], reinforcement learning was used to find music that could cause happiness in patients. Since the stimulus for the signals collected in the DEAP dataset consisted of music videos, this was used as the dataset for evaluation of the model, and the emotional wheel model was combined to train an exclusive music playlist that was suitable for different individuals. There are two main types of method based on DRL. The first is a model-free structure, which does not model the real world; an agent can only perform actions through certain policies in the real environment, wait for rewards and state transitions, and then update its behaviour policy based on this feedback. This process is repeated until the optimal policy is learned. The second is a model-based approach, in which a virtual world is constructed to simulate the states and interactions of the real environment, and strategies can be learned and planned in the virtual world. Yang et al. [42] proposed a reinforcement learning strategy to gradually learn the depth of the layers of a CNN and the connections between the layers. Li et al. [43] proposed a neural architecture search (NAS) framework based on DL to automatically design network structures. This framework searched for various possible combinations of neural network structures, obtained the optimal convolutional structure and verified the effectiveness of the resulting model, 97.74% for valence, 97.94% for arousal and 97.82% for dominance on DEAP. Since reinforcement learning mainly deals with sequence information, mining the contextual temporal information of EEG signals is a very important processing goal. Zhang et al. [44] proposed an unsupervised time-aware sampling network (TAS-Net) to detect EEG segments containing key information and to discard segments that were not directly related to emotion. Inspired by the physiological structure of the brain, Li et al. [45] proposed an end-to-end network architecture called a frontal lobe double duelling deep Q network (FLD3QN). By extracting the feature information from the left and right cerebral hemispheres and the left and right frontal lobe regions, the Q value was estimated using an RCNN, the fully connected layer and the obtained action probability was then input into the duelling architecture to obtain the final Q values for the current and the target Q networks.

Compared with single DL models, hybrid DL models have greater abilities in terms of feature representation. Their specific advantages are as follows. A hybrid model can combine the spatio-temporal characteristics to obtain global effective information, and in addition to retaining the effective features extracted by one model, the use of a

second model can overcome the limitations of a single structure. Through model fusion, the advantages of different models can be combined, and excellent results can be obtained. However, there are also some potential problems; for example, the complex internal structures of hybrid models impose certain requirements on the basic information contained in the input features. Hence, when constructing the input data, it is often necessary to retransform and combine the original feature dimensions in order to obtain and extract the most complete discriminative features. This means that the learning degree of the model for the training data is too consistent with the training individual itself, and the generalisation ability of the model for different individuals is ignored. In other words, in addition to the problem of improving the recognition accuracy of a single subject, the issue of how to improve the accuracy across subjects is also an important aspect of research on EEG emotion recognition.

3.4 Domain adaptation-based DL models

Domain adaptation (DA)-based DL models mainly focus on the deviation between the features of different individuals from the perspective of cross-individual recognition, and aim to maximise this difference through the use of domain adversarial, domain discriminator and adversarial losses. There are non-negligible differences between the EEGs of different individuals, and a simple transfer without adaptation may lead to misjudgement of the pre-trained model, resulting in poor classification performance for the target subject. The idea of deep DA was developed to address the issue of subject bias between the source domain (training samples) and the target domain (test samples), and to enable the recognition results to be better extracted by reverse adjusting the model parameters.

Lan et al. [46] used the DA method and compared the results on two public datasets, DEAP and SEED; their experiment proved the effectiveness of DA across datasets, improving the accuracy significantly by 7.25–13.40% compared to the baseline accuracy where no domain adaptation technique is used. Li et al. [47] increased the potential similarity of representation between the source and target domains by minimising the classification error to optimise the model performance. Wang et al. [74] proposed a DA symmetric and positive definite matrix network (daSPDnet). In the forward propagation process, the source data and target data represented by a symmetric and positive definite (SPD) matrix were input into the domain adaptive feature extractor with shared weights to obtain a low-dimensional representation, and the classifier output was calculated by prototype learning. The set of SPD matrices is defined as:

$$P(n) = \{P \in \mathbb{R}^{n \times n} | P^T = P, x^T P x > 0, \forall x \in \mathbb{R}^n\} \quad (7)$$

$n \times n$ symmetric matrices form a differentiable Riemannian manifold, and the distance between two points $P_1, P_2 \in P(n)$ can be accurately measured by the geodesic distances. Through backpropagation and adversarial training of the domain discriminator, the marginal distribution differences between the source and target representations were reduced. Bao et al. [48] proposed a two-level domain adaptive neural network in which the first-level domain adaptive network was composed of a feature extractor and a maximum mean difference (MMD) algorithm, with the aim of reducing the difference in the feature distribution between domains, and the second-level domain adaptive network consisted of a domain adversarial neural network (DANN) with a feature extractor and a domain discriminator. MMD distance can effectively measure the distance between distributions, and can be expressed by:

$$\begin{aligned} L_{\text{MMD}}(X_S, X_T) = & \frac{1}{n_S^2} \sum_{i,j=0}^{n_S} K(X_S^{(i)}, X_S^{(j)}) \\ & - \frac{1}{n_S n_T} \sum_{i,j=0}^{n_S, n_T} K(X_S^{(i)}, X_T^{(j)}) \\ & + \frac{1}{n_T^2} \sum_{i,j=0}^{n_T} K(X_T^{(i)}, X_T^{(j)}) \end{aligned} \quad (8)$$

where $X_S \sim x_i^S, i = 1 \cdots n_S$ is the data sample of the source domain, and $X_T \sim x_i^T, i = 1 \cdots n_T$ is the data sample of the target domain. n_S, n_T represent the number of samples in the source and target domains, respectively, and $K(\cdot, \cdot)$ is a linear combination of multiple radial basis function (RBF) kernels.

After processing by the two-level domain adaptive network, the distributions of the source and target domains became similar, and good classification performance was obtained. Zhao et al. [84] proposed a plug-and-play DA method with a focus on practicability, in which the sample labelling time was controlled to within one minute while the accuracy was maintained. He et al. [49] used temporal convolutional networks (TCNs), containing three-layer causal dilated convolution and three residual blocks, to extract feature information to preserve temporal characteristics; the encoded features were then input into the model based on an adversarial discriminative DA algorithm. Li et al. [50] proposed a dynamic DA algorithm that could realise a difference comparison from coarse-grained to fine-grained. To address the issue of neglected transferable differences in EEG signals, Li et al. [51] proposed a transposition multi-layer perceptron (TMLP) combined with a transposition multi-layer perceptron and sample-reweighted domain adaptive neural network (SRDANN) feature learning framework, which focused on sample

information with strong transfer. Guo et al. [52] also focused on spatio-temporal feature information and proposed a multi-source domain adaptation model with a spatio-temporal feature extractor (MSDA-SFE). The accuracy is respectively promoted by at least 2% and 3% in cross-subject and cross-session experiments compared with the best result of comparison methods (HVF₂N-DBR [53] and MS-MDA [54]). Wang et al. [55] proposed a multi-modal domain adaptive variational autoencoder (MMDA-VAE) to extract and learn the shared domain latent representation of EEG and eye movement signals.

DA is an important method that can be applied to EEG emotion recognition tasks. Although the accuracy of methods based on CNNs, RNNs, GNNs and Transformers is high in subject-dependent experiments, it is necessary to combine DA methods to achieve good recognition results in subject-independent experiments, as fine-tuning is needed to help make up for the differences between individuals. Structures such as discriminators must be added for reverse parameter fine-tuning to obtain accurate classification results in the target domain after these network models have been applied. To solve this problem, Li et al. [56] proposed an improved BiDANN-S model based on the asymmetric responses of the left and right hemispheres to emotion, reducing the domain difference between the source and target domains. Li et al. [57] added a domain discriminator to compensate for the difference between the training and testing samples to prevent overfitting. A cross-attention-based dilated causal convolutional neural network with domain discriminator (CADD-DCCNN) for multi-view EEG-based emotion recognition is proposed to minimise individual differences and automatically learn more discriminative emotion-related features by Li et al. [58] to deal with some limitations, including redundant and meaningless time frames and channels, as well as inter- and intra-individual differences in EEG signals from different subjects. Concerned about much fine-grained information, Li et al. [59] proposed a method called graph-based unsupervised subdomain adaptation (Gusa), which simultaneously aligns the distribution between the source and target domains in a fine-grained way from both the channel and emotion subdomains.

The models described above can effectively handle the problem of dealing with the mismatch in the feature distribution between the source and target domains, and can eliminate domain differences, extract the features common to different subjects as far as possible, and retain the discriminative information of different individuals to improve the generalisation ability and robustness of the models.

4 Datasets and comparative experiments

4.1 Datasets

Datasets collected based on physiological signals are typically multi-modal datasets. The following section presents a selection of EEG datasets that are frequently employed in the fields of EEG signal processing and brain–computer interface research, particularly in the context of emotion recognition.

1. DEAP [60]: This is a dataset for human emotional state analysis that contains data from 32 brain electrode channels and other physiological signals collected from 32 subjects while they were watching 40 one-minute music videos. For 22 of the subjects, one-minute videos of their faces were also recorded. After viewing each stimulus video, the subjects rated it in terms of arousal, valence, like/dislike, dominance and familiarity. When used for algorithm comparison experiments, most methods will customise the threshold according to the rating range, and change the indicators such as valence, arousal and dominance into binary or triple classification, and can also be divided into four emotional categories according to the VA space. Four discrete emotional categories are considered: Happiness (positive/high arousal), relaxation (positive/low arousal), boredom (negative/low arousal) and panic (negative/high arousal).
2. SEED [61]: A total of 15 subjects watched 15 movie clips in Chinese with a duration of approximately four minutes, which evoked positive, negative and neutral emotions in the viewers. In this experiment, EEG signals from 62 channels were collected from the participants. The sampling frequency was reduced to 200 Hz, and noise was removed by bandpass filtering from 0.3 to 50 Hz. The 15 subjects were required to watch each movie three times in total, at intervals of a week or more, meaning that the SEED dataset can also be applied to cross-session experimental research. The first signal collected from each individual can achieve higher recognition accuracy than the last two.
3. SEED-IV [68]: The experimental settings of this dataset are basically consistent with those of SEED. EEG signals were collected at the same time as eye movement signals, while participants were watching stimulus videos corresponding to four categories of emotion.
4. MAHNOB-HCI [113]: This dataset is representative of a multimodal dataset, comprising responses to emotional stimuli, and is intended for use in studies of emotion recognition and implicit labelling. It is recorded synchronously, encompassing facial video, eye gaze data and peripheral/central nervous system physiological signals. This enables multimodal emotion studies with feature-level and decision-level fusion.
5. DECAF [114]: The deeper EEG signals were extracted through the use of magnetoencephalogram (MEG), while near-infrared (NIR) facial videos of the subjects were also recorded.
6. DREAMER [75]: All signals in this dataset were captured with portable, wearable, wireless, low-cost devices. A total of 23 subjects were asked to complete a self-assessment after viewing 18 videos belonging to the stimuli.
7. MPED [99]: This dataset records a total of four physiological signals, namely EEG, galvanic skin response (GSR), respiration pattern (RSP) and electrocardiogram (ECG), and categorises emotion types into seven categories.
8. CAS-THU [115]: The dataset was obtained by collecting EEG signals from 30 participants who viewed stimulus video clips corresponding to each of the eight emotions, with two film clips for each emotion, which is conducive to the study of coarse- and fine-grained emotion recognition.
9. ASCERTAIN [116]: The objective is to acquire data using commercial sensors and to conduct a study of the nonlinear physiological correlations between emotion and implicit personality.
10. SEED-V [117]: The experimental settings of this dataset are largely aligned with those employed in SEED. In the self-assessment section, subjects were requested to rate the stimulus effect of the film on a scale of 5 indicating the optimal stimulus effect and 0 indicating no sensation that should be stimulated. The self-assessment settings captured in this dataset are more reasonable and therefore more reliable.
11. AMIGOS [118]: A multimodal research dataset was constructed by the researchers, comprising data on individual and group emotions, personality traits and states of mind. In the experiment conducted with an individual audience, 40 participants viewed 16 short videos evoking emotional responses. In addition to recording EEG, ECG and GSR data, high-definition videos of the participants' heads were also captured, along with RGB and depth whole-body videos.

12. SEED-GER/FRA [119]: The experimental settings of this dataset are essentially aligned with those of SEED, with minor discrepancies. The dataset records EEG and eye movement signals from German and French individuals, which can be employed to evaluate the subject-independent experimental impact of research methodologies on cross-cultural contexts.

The aforementioned datasets can be employed in the training of models for the recognition of emotions expressed through EEG. Moreover, multimodal datasets such as DEAP, MAHNOB-HCI, MPED and AMIGOS can be employed in the context of multimodal emotion recognition tasks. Moreover, SEED and SEED-IV can be employed as research datasets for cross-session experiments, whereas SEED-GER/FRA can be utilised in cross-cultural experiments. Furthermore, SEED-GER/FRA can be utilised in cross-cultural experiments. DEAP and SEED are the most frequently utilised in research on brain-computer interfaces, and are frequently employed as benchmarks for comparative experiments between diverse methodologies [120].

4.2 Evaluation metrics

In the field of emotion recognition, standard evaluation metrics are used to evaluate the performance of a classification model and to measure its generalisation ability. According to the true class and predicted class of the sample, it is divided into true positive (TP), false positive (FP), true negative (TN) and false negative (FN).

1. Accuracy (ACC): This represents the proportion of correct prediction results compared to the whole sample, and is defined as follows:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

2. Standard deviation (STD): It is used to quantify the variability of the predictions around the mean, and is defined as follows:

$$STD = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (10)$$

where n represents the total number of predictions, x_i represents the value of the i th prediction and $\bar{x}$ denotes the mean of all predictions.

3. Precision (P): The probability of a positive sample being correctly identified as such is represented by Eq. (11):

$$P = \frac{TP}{TP + FP} \quad (11)$$

4. Recall (R): The proportion of true positives correctly identified out of all positives is measured, as demonstrated by Eq. (12):

$$R = \frac{TP}{TP + FN} \quad (12)$$

5. F1-Score: In order to achieve an equilibrium between precision and recall, a weighted harmonic average of P and R is employed. This metric is designated as F-Measure, alternatively known as F-Score, as illustrated in Eq. (13):

$$F_\beta = \frac{(1 + \beta^2) \times P \times R}{(\beta^2 \times P) + R} \quad (13)$$

When $\beta = 1$, the F1-score is sought, which is the reconciled average of P and R.

6. Precision-Recall (PR) Curve: It is a graphical tool for evaluating the performance of binary classification models, which plots the predictive precision and recall of the model at different confidence levels in a two-dimensional coordinate system based on actual labels.
7. Confusion Matrix (CM): This is a table used for the evaluation of a classification model, demonstrating the correlation between the model's predictions across different categories and the actual labels. Furthermore, the confusion matrix allows for the accurate identification of instances where the model incorrectly classifies positive examples as negative or vice versa, or alternatively, where negative examples are incorrectly classified as positive.
8. Receiver Operating Characteristic-Area Under Curve (ROC-AUC): The ROC curve is a two-dimensional coordinate plot in which the horizontal axis represents the false-positive rate (FPR) and the vertical axis represents the true-positive rate (TPR), also known as recall or sensitivity. AUC is the abbreviation for area under the ROC curve, which is the integral value under the curve. The value of AUC ranges from 0 to 1, with a value closer to 1 indicating a superior model performance and a value closer to 0.5 indicating a suboptimal model performance.

Among all the evaluation metrics, accuracy and standard deviation are usually used as the main evaluation metrics, which can reflect both the recognition effect and the stability of the model. Furthermore, confusion matrices are also frequently employed as a primary evaluation metric for models, facilitating the analysis and enhancement of

classification model performance by illustrating misclassified samples across different categories.

4.3 Experimental methods and analysis

4.3.1 Subject-dependent experiments

In subject-dependent experiments, the sample features fed to the neural network for classification are not individually discriminative, and the EEG signals of each subject are present in both the training and test datasets. In general, 80% of the samples in the dataset are used as training data, with the remaining 20% as testing data, and the evaluation results are obtained by using fivefold or tenfold cross-validation.

DEAP and SEED are often used as benchmark datasets to compare the results, in addition to which the results of the methods using SEED-IV, DREAMER and MPED are also summarised. Table 3 shows the comparison results of existing methods proposed by different basic networks based on deep learning. State-of-the-art classification results are shown in bold under each model. The column entitled 'Protocol' shows the training strategy used in the corresponding method, and the ratio represents the division of the training and test sets.

Of the EEG emotion recognition algorithms shown in Table 3, the state-of-the-art methods are SFE-Net [16] for the SEED dataset and the scheme in Ref. [101] for DEAP dataset. These two methods preserve the spatial, temporal and channel characteristics of EEG signals as far as possible during feature extraction, and prove that 3D-CNN and Transformer encoders have a strong ability to retain and learn important feature information. The sparse DGCNN [13] is the best method based on a graph neural network, which also reflects the important reference value of sparse spatial topological matrix mapping in tasks related to EEG data. Figure 8 shows the increases in recognition effectiveness of subject-dependent experiments on the DEAP and SEED datasets over recent years. It can be seen from the figure that the highest ACC values for the two datasets were both reached in 2021; since then, the accuracy of DEAP has remained stable at above 90%, while the accuracy of SEED has remained stable at above 88%. In recent years, subject-dependent experiments have achieved good results.

The DE features of the SEED dataset can be divided into five frequency bands, and a comparison of the experimental results for each frequency band is important as a reference, as shown in Table 4. The input features of all methods are DE. It can be seen that the Sparse DGCNN [13] has the best results in each frequency band; however, this method was not verified in the delta frequency band, as the study showed that the delta frequency band had a low response intensity to EEG emotional fluctuations. Removing the feature processing and learning of the delta band can therefore improve the recognition performance. Although the classical subject-dependent experiment performs well in emotion recognition, its flexible application is limited by the risk of overfitting and the dependence on the amount of data, especially in regard to verification of the generalisation performance of the model.

The confusion matrices of subject-dependent experiments of representing methods are shown in Fig. 9, where the element located in the i th row and j th column shows the percentage of those samples belonging to the class i while predicted as the class j . Diagonal elements mean the accuracies of each class and others mean the confusion.

Representative methods and experimental results for the visualisation of misclassified samples on DEAP, SEED and SEED-IV are shown. (a–d) shows the results of the four dimensions of valence, arousal, dominance and VA space on DEAP, where 'H' stands for 'high' and 'L' for 'low', (e) shows the results of BiHDM on SEED, and (f) shows the results on SEED-IV, using the same methodology as the experiments on SEED, where 'NEG' stands for 'negative', 'NEU' for 'neural', 'POS' for 'positive', 'FEA' for 'far' and 'HAP' for 'happy'. Combining the confusion matrices of these representative methods shows that positive types are easily misclassified into negative types, while samples belonging to the sad and fear types of negative types are easily misclassified into each other's categories.

4.3.2 Subject-independent experiments

EEG subject-independent experiment refers to the use of cross-subject data to train and evaluate models in emotion recognition research. Different from the traditional training and testing methods based on a single subject's data, subject-independent experiments usually adopt the way of setting aside a validation set or cross-validation to ensure that the training set and test set come from different subjects. In the subject-independent experiment, the participant's dataset is divided into a training set and a test set.

Table 3 Comparison of subject-dependent experimental results

Categories	Methods	Input features	Training details		
			Loss	Optimizer	Protocol
Single	CNN	EEG with rearrangement channels	Categorical CEL	Adadelata [121]	Fivefold
		2D array:	Categorical CEL	SGD	LOO
		40×101			
		3D tensor	CEL	Adam	Fivefold
		DE	AM [122]	SWA [123]	Tenfold
	RNN	3D tensor:	Weighted CEL	BP	Fivefold
		$8 \times 8 \times 9$			
		Raw EEG	CEL	Adam	Tenfold
		DE	CEL		LOO
		DE	CEL + discriminator	SGD	SEED:9:6 SEED-IV:16:8 MPED:12:7
Attention-based	GNN	DE	KL divergence + sparsity		SEED:9:6
					MPED:12:7
		3D	Contextual loss + Adjacency loss	Adam	DEAP:tenfold
		spatial-spectral features			DREAMER:
		DE			18-fold
	CNN	Region features	CEL + KL divergence	Adam	SEED:9:6 SEED-IV:16:8
		SF	CEL + Entropy minimization loss	Adam	SEED:9:6 SEED-IV:16:8
		DE			Tenfold
			CEL + KL divergence + L1norm	GD	SEED:9:6
					MPED:12:7 DREAMER:
TF		3D tensor:	CEL	Adam	LOO
		$C \times V \times H$			Fivefold
		Raw EEG			
		EEG without baseline	Margin	Adam	7:3
		Raw EEG		Adam	Tenfold
					4:1

Table 3 continued

Categories	Methods	Input features	Training details		
			Loss	Optimizer	Protocol
Hybrid	CRNN	Raw EEG			Fourfold
		EEG without baseline	CEL	Adam	Tenfold
		DE			
		4D tensor: $h \times w \times d \times 2 \times T$	CEL	Adam	Tenfold
		4D tensor: $h \times w \times 2f \times 2 \times T$		Adam	Fivefold
	GCN	DE			
		SEED:DE DREAMER: PSD	CEL + regularization	BP	Tenfold
		DE	NLL	Adam	SEED-9:6 DREAMER: LOO
		Five frequency bands	CEL	GD	Tenfold
					SEED-IV:CS
		SEED:DE		AdamW	SEED-V:CS
		SEED-IV:DE			MPED:12:7
		SEED-V:DE			SEED:3:2
		MPED:STFT			SEED-IV:2:1
		DE + PSD	CEL + regularization		SEED-V:threefold
		DE	CEL	Adam	MPED:3:1
		DEAP:PSD	CEL + regularization	BP	Fivefold
		SEED:DE + PSD DREAMER:			Tenfold
		DE + PSD	CEL + L1norm	BP	LOO
		STFT			
	GNN + LSTM	DE			12:7
		Raw EEG	CEL	Adam	Tenfold
		Parallel channel + position reconstruction	CEL	Adam	Fivefold
		EEG sequence			4:1
		DE			
	DRL	Raw EEG	CEL	Adam	Tenfold
		Raw EEG	Policy gradient	Adam	tenfold
			MSE	GD	Tenfold

Table 3 continued

Categories		Methods	Input features	Training details		Protocol
				Loss	Optimizer	
Domain adaptation		[58], 2024	Five frequency bands	DD	Adam	LOO
		[56], 2018	DE	CEL + DD	SGD	9:6
		[57], 2019	DE	CEL + DD	SGD	9:6
Results: ACC $\pm$ STD (%)						
Categories	DEAP		SEED	SEED -IV	DREAMER	
	V	A			V	A
Single	CNN	77.98	73.98			
		81.41	73.36			
		91.94	92.49	99.19		
		95.02	95.29	88.84		
		96.28	96.62			
		96.65 $\pm$ 2.65	97.11 $\pm$ 2.01		95.55 $\pm$ 2.18	97.01 $\pm$ 2.74
	RNN		89.50 $\pm$ 7.63			
			93.12 $\pm$ 6.06	74.35 $\pm$ 14.09		40.34 $\pm$ 7.59
	GNN	57.73	58.12	94.30	63.38	67.18
				94.24 $\pm$ 5.95	79.37 $\pm$ 10.54	
				96.72 $\pm$ 4.58	82.78 $\pm$ 11.93	
Attention-based	CNN	94.01	93.55			
	GNN			95.64 $\pm$ 5.08	92.82	93.09
	TF	92.86	93.34	96.28 $\pm$ 4.39	83.27 $\pm$ 8.37	40.40 $\pm$ 9.35
		98.76 $\pm$ 1.33	98.81 $\pm$ 1.46	88.08 $\pm$ 6.32	98.59 $\pm$ 1.38	98.61 $\pm$ 1.34
		99.40	99.10			

Table 3 (continued)

Categories	Results: ACC $\pm$ STD (%)						
	DEAP		SEED		DREAMER		MPED
	V	A	SEED	SEED -IV	V	A	
Hybrid	CRNN	88.30					
		90.10					
		90.80 $\pm$ 3.08					
		93.72 $\pm$ 3.21					
		94.22 $\pm$ 2.61					
			94.74 $\pm$ 2.32				
			96.10 $\pm$ 2.61				
			90.22 $\pm$ 3.67				
			90.40 $\pm$ 8.49				
			93.19 $\pm$ 1.54				
		94.53 $\pm$ 1.02	94.91 $\pm$ 1.05				
				80.93 $\pm$ 1.19			
					86.23 $\pm$ 12.29		
					94.57 $\pm$ 1.05		
						84.54 $\pm$ 10.18	
						94.82 $\pm$ 1.03	
							43.56 $\pm$
							8.21
							40.54 $\pm$
							7.02
			96.93 $\pm$ 5.11	82.24 $\pm$ 14.85			
		95.91 $\pm$ 3.27	97.31 $\pm$ 1.47	89.97 $\pm$ 2.85			
		94.97 $\pm$ 2.50	98.98 $\pm$ 1.50				
		96.18 $\pm$ 2.54	99.01 $\pm$ 2.67		95.92	94.73	
							40.69 $\pm$ 11.12
	GNN + LSTM						
	CNN		93.83				
	+		95.30				
	TF		97.64		96.80	97.74	
		98.27	98.53	91.90 $\pm$ 5.43			
			98.47 $\pm$ 1.72				
	DRL	97.74 $\pm$ 1.02	97.94 $\pm$ 1.04		96.29 $\pm$ 3.82	96.62 $\pm$ 3.52	
		98.35	98.17		92.60	92.30	
		90.97 $\pm$ 13.96	92.42 $\pm$ 12.72				
Domain adaptation			87.40 $\pm$ 1.80				
			92.38 $\pm$ 7.04				
			93.38 $\pm$ 5.96				

[†] V, valence; A, arousal; CEL, cross-entropy loss; SGD, stochastic gradient descent; LOO, leave-one-out; SF, selected features; AM, additional margin softmax loss; SWA, stochastic weight averaging; BP, backpropagation; KL, Kullback–Leibler; STFT, short time Fourier transform; NLL, negative log likelihood; GD, gradient descent; CS, cross-session; MSE, mean square error; DD, domain discriminator

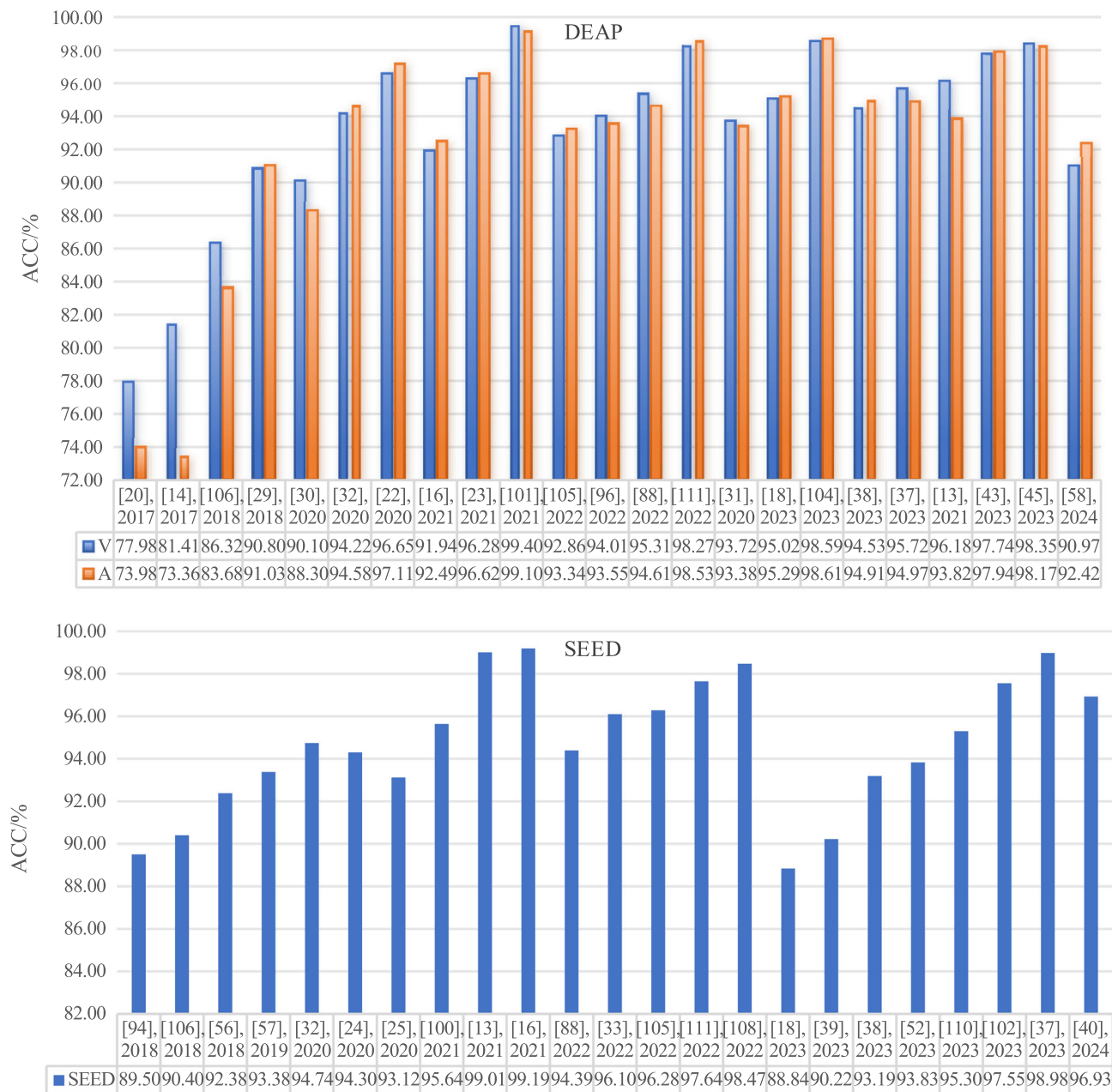


Fig. 8 The growth trends of ACC for subject-dependent experiments on DEAP and SEED

For each test set, the data from one of the participants are retained as test data, while the data from the remaining participants is used as training data to construct a classification or recognition model.

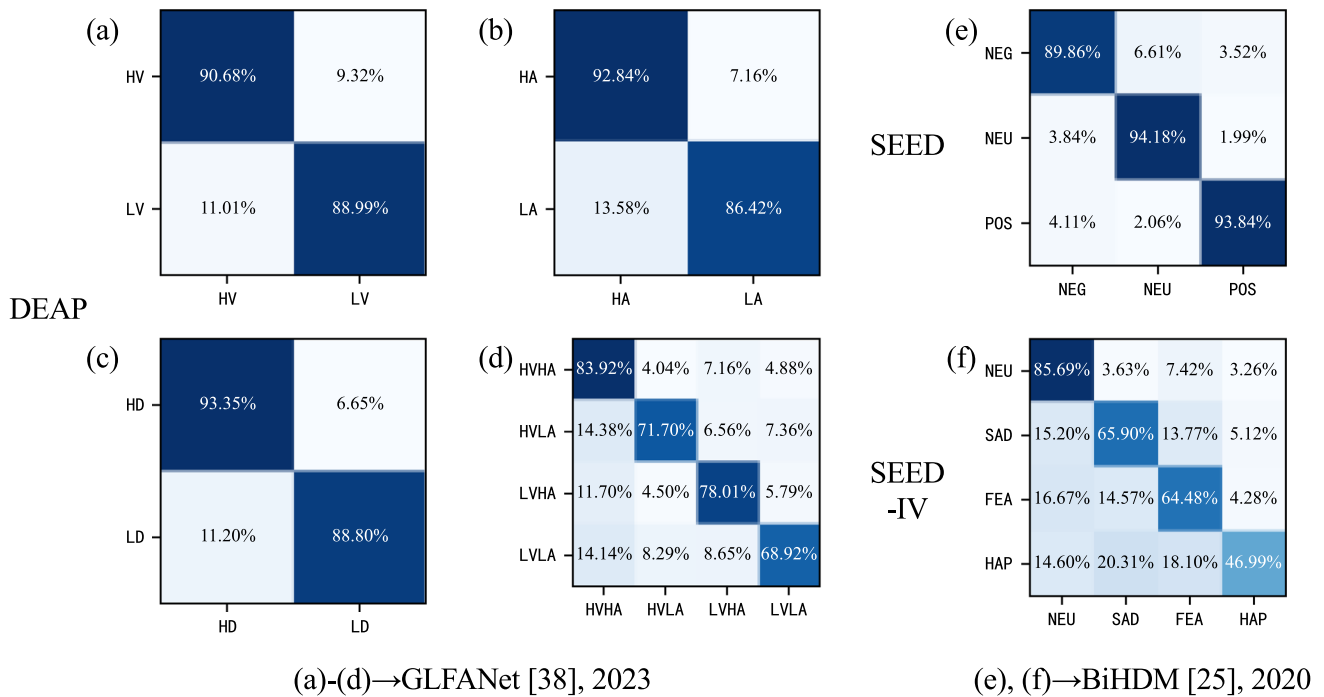
The primary purpose of the subject-independent experiment is to compare the best trained model. Due to the large differences between the EEG signals of different individuals, this comparison can be used to derive the generalisation ability of the model. In recent years, some studies have involved subject-independent experiments, also known as cross-subject experiments, to realise the rapid application of emotion recognition in brain–computer interfaces.

Table 5 shows the results of the current part of the subject-independent experiments, which reflect the better logical generalisation ability and the ability to optimise the data bias for individual differences in domain adaptation. It is highly likely to promote the accuracy of emotion recognition for unknown subjects in practical applications. Furthermore, subject-independent experiments offer a means of objectively assessing and comparing the different methods, which is conducive to the continuous development of the field.

This experimental method was originally proposed by Song et al. [106], and the LOSO method was used to

Table 4 Comparison of subject-dependent experimental results of frequency bands for SEED

Method	Year	Results: ACC $\pm$ STD (%)					
		δ	θ	α	β	γ	All
STRNN [94]	2018	80.90 $\pm$ 12.27	83.35 $\pm$ 9.15	82.69 $\pm$ 12.99	83.41 $\pm$ 10.16	69.61 $\pm$ 15.65	89.50 $\pm$ 7.63
CSGNN [39]	2023	54.67 $\pm$ 11.28	56.67 $\pm$ 12.38	62.74 $\pm$ 10.94	83.63 $\pm$ 8.39	93.63 $\pm$ 4.33	90.22 $\pm$ 3.67
DGCNN [106]	2020	74.25 $\pm$ 11.42	71.52 $\pm$ 5.99	74.43 $\pm$ 12.16	83.65 $\pm$ 10.17	85.73 $\pm$ 10.64	90.40 $\pm$ 8.49
BiDANN [56]	2018	76.97 $\pm$ 10.95	75.56 $\pm$ 7.88	81.03 $\pm$ 11.74	89.65 $\pm$ 9.59	88.64 $\pm$ 9.46	92.38 $\pm$ 7.04
R2G-STNN [57]	2019	77.76 $\pm$ 9.92	76.17 $\pm$ 7.43	82.30 $\pm$ 10.21	88.35 $\pm$ 10.52	88.90 $\pm$ 9.97	93.38 $\pm$ 5.96
DCoT [109]	2022	86.91	87.77	88.65	89.38	90.75	93.83
RGNN [28]	2020	76.17 $\pm$ 7.91	72.26 $\pm$ 7.25	75.33 $\pm$ 8.85	84.25 $\pm$ 12.54	89.23 $\pm$ 8.90	94.24 $\pm$ 5.95
V-IAG [100]	2021	81.14 $\pm$ 9.46	82.37 $\pm$ 7.44	84.51 $\pm$ 9.68	92.15 $\pm$ 8.90	92.96 $\pm$ 6.19	95.64 $\pm$ 5.08
PGCN [40]	2024	79.62 $\pm$ 10.53	83.62 $\pm$ 6.91	83.74 $\pm$ 9.57	92.33 $\pm$ 8.66	93.05 $\pm$ 5.78	96.93 $\pm$ 5.11
Sparse DGCNN [13]	2021		97.30	97.22	98.43	98.34	98.53

**Fig. 9** The confusion matrices of subject-dependent experiments

separate the source and target subjects. The LOSO method is a cross-validation technique which is used to evaluate and validate the performance of EEG classification or recognition models. It is often used in cross-subject experiments to address individual differences and generalisation performance. By evaluating each participant's data as test data, the subject-independent experiment can more accurately assess model performance and infer the model's ability to generalise and make predictions on new participants.

These results in Table 5 also demonstrate the importance of domain adaptation in EEG emotion recognition tasks. CADD-DCCNN [58] gives the best performance on DEAP, whereas DBGCN [87] gives the best performance on the SEED and SEED-IV datasets. The success of CADD-DCCNN can be largely attributed to its utilisation of multi-view learning combined with attention mechanisms to effectively select emotion-related channels and time frames. Additionally, the model leverages the ability of dilated causal convolutional neural networks to extract temporal information from multi-view features. DBGCN

Table 5 Comparison of subject-independent experimental results for DEAP, SEED and SEED-IV, DREAMER

Categories	Methods	Results: ACC ± STD(%)					
		DEAP		SEED		SEED-IV	
		V	A	D		V	A
Single	RNN GNN	[25], 2020				69.03 ± 8.66	
		[26], 2023	57.73	58.35	61.69	85.40 ± 7.53	61.84
Attention -based	CNN GNN	[28], 2020				73.84 ± 8.02	67.98
		[97], 2024	69.40 ± 7.04	70.04 ± 10.13	70.98 ± 13.48	92.12 ± 3.32	78.98 ± 5.17
	TF GCN	[100], 2021				88.38 ± 4.80	
		[103], 2022	66.51 ± 8.53	65.75 ± 8.51		79.95 ± 9.02	
Hybrid	GCN	[106], 2018				81.85	
		[39], 2023				84.59 ± 8.68	
		[40], 2024				73.69 ± 7.16	
		[35], 2023				76.96 ± 7.19	
		[13], 2021	61.93 ± 6.03	65.94 ± 8.80		88.67 ± 11.23	67.12
Domain adaptation	CNN + TF GNN + TF	[87], 2022				94.39 ± 3.23	
		[109], 2022				83.03 ± 5.70	
	DRL	[112], 2023				90.37 ± 6.18	75.26 ± 9.71
		[44], 2023	57.84	60.51	63.17	63.10	
	DREAMER	[51], 2023	62.88 ± 5.55	57.70 ± 7.23		81.04 ± 6.28	
		[56], 2018				84.14 ± 6.87	
		[57], 2019				84.16 ± 7.63	
		[49], 2022	64.33 ± 7.06	63.25 ± 4.62			63.69 ± 6.57
		[55], 2022				85.07 ± 11.81	
		[48], 2021				87.90 ± 6.13	
		[50], 2022				91.08 ± 7.70	81.58 ± 8.72
		[52], 2023	69.26	70.10		91.65 ± 2.91	73.92 ± 6.04
		[59], 2024				91.77 ± 5.91	75.12 ± 7.99
		[58], 2024	69.45 ± 5.60	70.50 ± 9.39		92.44 ± 6.16	

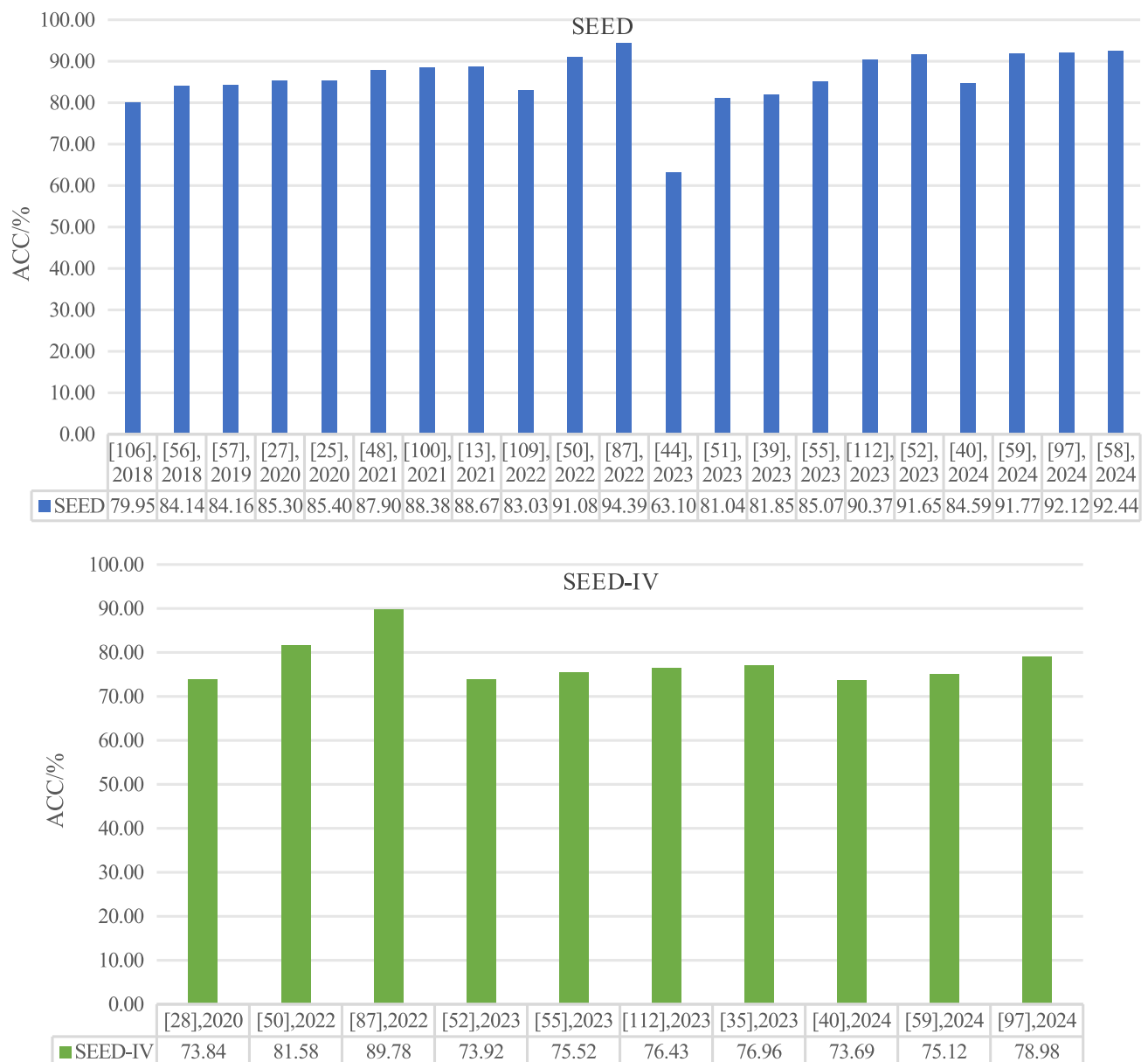


Fig. 10 The growth trends of ACC for subject-independent experiments on SEED and SEED-IV

combines the two-branch dynamic graph convolution and adapter-tuned transfer learning (AFTL) methods, and considers both the spatial characteristics and the differences between individuals. Excellent results are achieved in both the subject-dependent and subject-independent experiments: The accuracy of SEED is improved to 94.39%, and the accuracy of SEED-IV to 89.78%. It can also be seen from Table 5 that the results for the DEAP dataset are not as good as for SEED and SEED-IV. The reason for this may be that DEAP dataset only contains data from 32 EEG channels, and is missing some associated EEG information that helps in the extraction of common features of individual EEGs. STGATE [112]

achieves 77.44% in the valence dimension, 75.26% in the arousal dimension and 76.35% in the average value in both dimensions in DREAMER dataset, because the model can extract more useful information in the temporal and spatial dimensions.

Figure 10 shows the increases in recognition effectiveness of subject-independent experiments on the SEED and SEED-IV datasets over recent years. It can be seen from the figure that the highest ACC values for the two datasets were both reached in 2022; since then, the accuracy of SEED has remained stable at above 63%, while the accuracy of SEED-IV has remained stable at above 73%. In

recent years, subject-independent experiments have attracted the attention of more and more researchers.

Combining the results of subject-independent experiments on the above models, we can conclude the following research experiences. Firstly, such experiments can effectively mitigate the impact of subject differences on the performance of emotion recognition algorithms. Given the considerable inter-subject variability in the structure and function of the human brain, it is frequently challenging to extrapolate a model trained on some acquired subjects to new subjects. Subject-independent experiments can assess the generalisation ability of the algorithm in a more objective manner through cross-subject training and testing, which is conducive to improving its robustness in practical applications. Secondly, subject-independent experiments can assist in identifying and eliminating the issue of “data bias” within the models. The neural activity patterns of specific subjects may be overfitted, which may result in the models being difficult to generalise. Subject-independent experiments can assist researchers in identifying and solving this problem, thereby improving the general applicability of the algorithm. Furthermore, the outcomes of the subject-independent experiments are more aligned with real-world applications, thereby providing a reliable foundation for the transformation of EEG emotion recognition technology into a functional system. This is conducive to promoting technological innovation and industrialisation in this field, and accelerating the deep integration of artificial intelligence and emotion computing.

Overall, the EEG subject-independent experiments provide an important research idea for exploring more robust and universal emotion recognition algorithms, which is of great significance for promoting the practical application of brain–computer interface technology in the fields of intelligent interaction and mental health.

5 Discussion

5.1 Applications and trends

EEG emotion recognition systems have been used in a range of different fields. EEG-based brain–computer interface (BCI) adds to the monitoring processes, for example via enabling systems to be adaptive or sending alert signals [124]. Real-time fatigue monitoring [125] for drivers on the road is one important application of such recognition systems. The EEG signals collected by wearable devices are used to detect whether the driver is in a negative mood or a fatigued state, in order to better monitor the problems encountered in the driving process and ensure driving safety. In the field of entertainment, real-time

detection of the emotional changes of game players can be carried out [126] and the players’ choices in the game scenarios can be inferred, meaning that the quality and value of such games can be improved, and the emotional experience related to the human–computer interaction can be enhanced. Sethi et al. [127] uses Neurosky Mindwave headset, a non-invasive, EEG-based BCI to measure the attention level of users. They present an experiment to study the impact of real-time attention-based neurofeedback in improving the performance and average attention of users while performing a reading comprehension task. Each participant is provided with two different reading comprehension tasks, one without neurofeedback and other with the attention-based neurofeedback. The experiment results indicate an improvement in the average attention as well as improvement in the test scores of the users.

In the field of emotion recognition, multi-modal information can be used to complement data, effectively reduce the redundancy of information, represent features better and enhance the robustness of a system. Fusion between EEG data and other physiological signals or video, image and other modalities is a direction that researchers will continue to pay attention to in future. The fusion of EEG information with face recordings can improve the accuracy of individual expression recognition [86]. This makes human–computer interaction more convenient, and allows an intelligent personal assistant to accurately identify the user’s emotions, thereby improving the level of communication with humans. Functions such as multimedia content recommendation can also be improved to a certain extent, to take into account the real needs of users. In addition to its use in daily life, it can also be applied in the field of medical assistance to meet the special needs of hearing people who lack auditory perception channels, and deaf mute, paralysed and other patients. It can be used to explore their deviation of emotional understanding, facilitate timely communication with patients, monitor their mental health and help autistic children perceive others’ feelings. It can even be applied in public prosecution departments; criminal suspects may hide their true emotions during interrogation, and it is necessary to combine EEG data with other physiological signals to determine whether the content of a confession expresses the true thoughts of the individual. In order to make the use of BCI applications more efficient and further integrate the possibilities into our daily tasks, BCIs could be integrated with augmented reality which would create new dimensions of user experience and practicality [60].

5.2 Unresolved problems

- 1) Lack of large datasets and limitations in applications: A shortage of large datasets is one of the problems

that need to be solved. Today's datasets are mostly collected in the laboratory, and more data collection in the field is needed for further research in future. At the same time, few-shot datasets have significant limitations, and overfitting is common in the training process of some deep models. EEG datasets are typically comprised of a limited number of individual subjects, often only a few dozen. The restricted scope of small datasets may result in a lack of variability and a potential inability to fully reflect the diversity of the population. The quality of the recorded signal has an impact on the subsequent analysis and interpretation of the results. Inconsistencies in environmental settings, sampling frequency, baseline duration of the data collection process, artefacts, noise and inconsistencies in the data can impact the reliability and validity of the results.

- 2) Shortcomings of different network frameworks and difficulties of DL applying: Each deep learning method has particular advantages and disadvantages. A CNN can obtain the local features, while the self-attention mechanism in the Transformer can obtain global features, and the GNN can construct static or dynamic spatial and functional topology information and correlations. Deep networks based on RNNs and LSTM can extract time series information and causal connections from the context. The DA method can obtain the similarity features between subjects by reducing the sample differences between the source and target domains. The problem of how to integrate the advantages of multiple network structures into a complete network framework to realise end-to-end EEG emotion recognition remains to be solved. The use of overly complex network models can result in scenarios where subject-dependent recognition is subjected to excessive fitting and subject-independent recognition is underexposed, ultimately leading to suboptimal outcomes, such as [106].
- 3) Subject variability and hard improvements in cross-subject accuracy: The identification results from subject-independent experiments on datasets other than SEED are not satisfactory. The issue of how to improve the results of cross-individual experiments while ensuring the high accuracy of subject-dependent experiments has yet to be resolved. Individual differences between participants in a study can affect the results of a study and the generalisability of the results. This is particularly the case for subject-independent experiments, which are prone to unsatisfactory identification results if the number of individuals is too large or if there is too much disparity in characteristics between different individuals. For example, the accuracy of subject-

independent experiments for the validity of the DEAP dataset is usually no more than 70% [128].

5.3 Solutions

Some possible solutions to the problems described above are as follows:

1. Dataset collection: The continuous development of EEG equipment means that increasing numbers of commercial wearable acquisition devices are available. Hence, it is no longer necessary to acquire data in a laboratory environment. In order to apply EEG recognition tasks to daily life, researchers should consider collecting data in the field, and noise reduction processing of EEG signals could be carried out at the same time to reduce the influence of the external environment. In addition, the subjects' acceptance of the stimulus video should be fully considered when using self-filling questionnaires or other forms of self-evaluation. Before a formal signal acquisition experiment is carried out, it is necessary to refine the stimulus video that can best cause emotional fluctuations through the reflection of other participants, and to minimise the length of the videos to avoid fatigue to participants. Furthermore, ensuring the secure use of EEG-based BCI applications is a significant concern in users' everyday operation. In practice, the issuance of commands to the device at any given time should be unaffected by a specific temporal window. In the event that the user does not intend to utilise the device, it is imperative that the user is effectively disengaged from the device.
2. Model optimisation: In subject-dependent experiments, deep learning methods based on graph convolution can achieve high accuracy, and can retain and process the correlation information between EEG electrode spaciality. Temporal attention should be added to preserve more information within and between seconds. In the final training stage, a discriminator can be used to fine-tune the parameters, reduce the sample differences between individuals and enhance both the generalisation ability and the robustness of the model. In conclusion, the process of model optimisation should fully consider the feature states of EEG signals, extract the channel, spatial and temporal information of EEG signals from the perspective of subject-dependence, apply the idea of transfer learning to extract the domain-invariant features between individuals and reduce the data heterogeneity by combining with federated learning from the perspective of subject-independence.

3. **Common feature extraction:** The EEG signals of different individuals are significantly different. In addition to the different functional connection areas of brain regions, the transmission paths of different emotion fluctuations between electrode positions also differ. The issue of how to find common characteristics between different individuals is one of the keys to solve the differences between individuals. In network modelling, the characteristic representation of emotion EEG is simulated and established based on the correlations between EEG characteristics and channels, and the common features belonging to the same emotion with discrimination are dug deeply. The problem of different electrode channels collected from different databases can be solved by converting raw EEG signals into EEG topography maps. In the process of generating the EEG topography map, the interpolation algorithm is used to compensate for the positional data of the unset sensor nodes, which can compensate for the defect of too few EEG electrodes collected in some databases.

6 Conclusions

Deep learning-based EEG emotion recognition has developed rapidly and is widely used in various fields. Based on the basic theory of EEG emotion recognition, this paper divides the current deep learning models for EEG emotion recognition into four categories: Single, attention-based, hybrid and DA for detailed analysis. Then, subject-dependent and subject-independent experiments are analysed on the public EEG datasets, and it is concluded that subject-independent experiments still had more challenges but more application prospects. In view of the current limitations in dataset application, model optimization and subject variability, feasible solutions are proposed, and the future development trend of this field is prospected. This paper comprehensively and systematically summarises the key contents of EEG emotion recognition based on deep learning, which is helpful for readers to deeply understand the research status and future direction in this field, and is of great significance to promote the innovation and development of EEG emotion recognition technology.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interests The authors declare that they have no competing interests.

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